

Global Spherical Shells on Minimal Surfaces of Class VII

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September 24, 2026

Abstract

We prove the Global Spherical Shell conjecture: every connected minimal compact complex surface of class VII with positive second Betti number contains a global spherical shell. This is a holomorphically embedded neighborhood of the standard three-sphere in $\mathbb{C}^2 \setminus \{0\}$ whose complement is connected.

1 Introduction

A compact complex surface is of *class VII* if its first Betti number is one and its Kodaira dimension is $-\infty$. The classification of these non-Kähler surfaces divides according to the second Betti number. In the minimal case with $b_2 = 0$, work of Bogomolov, Li–Yau–Zheng, and Teleman leads to the classification by Hopf and Inoue surfaces [5, 23, 28]. The positive- b_2 case is the setting of the Global Spherical Shell conjecture.

A *global spherical shell* in a complex surface is a holomorphically embedded neighborhood of the unit three-sphere in $\mathbb{C}^2 \setminus \{0\}$ whose complement is connected. Kato introduced this structure [20]. Nakamura’s classification program includes the conjecture that every minimal class VII surface with positive second Betti number contains such a shell [25, Working Hypothesis 5.5]; see also [26, 30]. We prove this assertion.

Theorem 1.1 (Global Spherical Shell conjecture). *Let X be a connected compact complex surface with*

$$b_1(X) = 1, \quad b_2(X) > 0, \quad \kappa(X) = -\infty.$$

Assume that X is minimal: it contains no smooth rational curve of self-intersection -1 . Then there are an open neighborhood U of

$$S^3 = \{z \in \mathbb{C}^2 : |z| = 1\} \quad \text{in } \mathbb{C}^2 \setminus \{0\},$$

an open subset $\Sigma \subset X$, and a biholomorphism $\phi : U \rightarrow \Sigma$ such that $X \setminus \Sigma$ is connected.

Kato’s deformation theorem then identifies the deformation behavior and smooth topology of every surface in this range. A primary Hopf surface is a compact quotient of $\mathbb{C}^2 \setminus \{0\}$ by the infinite cyclic group generated by a holomorphic contraction.

Corollary 1.2 (Deformation and smooth topology). *Let X satisfy the hypotheses of Theorem 1.1, and put $b = b_2(X)$. There is a small one-parameter deformation $\mathcal{X} \rightarrow \Delta$ over a disk about 0, with $\mathcal{X}_0 \simeq X$, such that every nonzero fiber is obtained from a primary Hopf surface by exactly b successive point blowups. The centers may be infinitely near. Moreover,*

$$\pi_1(X) \cong \mathbb{Z}, \quad X \cong_{\text{or}} (S^1 \times S^3) \# b \overline{\mathbb{CP}}^2,$$

where the second is an orientation-preserving diffeomorphism for the complex orientation of X .

The theorem also removes the possible class VII exception in the geography of aspherical complex surfaces studied by Albanese–Di Cerbo–Lombardi [1]. Here asphericity means that the topological universal cover is contractible.

Corollary 1.3 (A sharpened Euler–signature inequality). *Let S be a connected closed smooth complex surface whose topological universal cover is contractible. Write $\chi_{\text{top}}(S)$ for its Euler characteristic and $\sigma(S)$ for its signature in the complex orientation. Then*

$$\chi_{\text{top}}(S) \geq \frac{9}{5}|\sigma(S)|.$$

If $\chi_{\text{top}}(S) > 0$, then S is of general type; otherwise $\chi_{\text{top}}(S) = \sigma(S) = 0$.

The proofs of these consequences are given in Section 7.1. We turn to the geometric reductions and the analytic construction that prove the main theorem.

Curves, shells, and earlier approaches

The shell construction makes the positive- b_2 classification problem concrete: one modifies a ball and glues its boundary neighborhoods by a holomorphic embedding. Kato introduced the shell framework [20]; Dloussky developed its description by sequences of blowups and contracting germs [8]. The conjecture asks whether an arbitrary minimal class VII surface with positive second Betti number admits this geometry, even when no curves are initially known to exist.

Two curve criteria reduce the task. Enoki’s theorem handles the presence of a nonzero effective divisor of square zero [14]. In the remaining case, Dloussky, Oeljeklaus, and Toma prove that $b_2(X)$ rational curves imply a global spherical shell [11]. Throughout this paper a rational curve may be singular: its normalization is $\mathbb{P}^1$. These criteria shift the burden to producing sufficiently many rational curves on a surface whose intersection form is negative definite. A topological cohomology class alone does not provide an effective divisor, and an effective divisor on a noncompact cover need not have compact components.

Gauge theory has supplied major advances in this curve-production problem. Teleman proved the shell conjecture for $b_2 = 1$ and proved that a minimal class VII surface with $b_2 = 2$ contains a cycle of curves [29, 30]. The cycle conclusion is distinct from the general criterion requiring b_2 rational curves. Dloussky’s numerically anticanonical and later twisted-logarithmic-form criteria give further routes from additional geometric structure to a shell [9, 10]. Kurnosov and Spicer obtain the shell conclusion for $b_2 = 3$ when two distinct singular holomorphic foliations are present [22].

Infinite divisors on covers also occur in the deformation work of Dloussky and Teleman [12]. For families of surfaces with shells, they describe lifted exceptional divisors whose limits can be infinite sums of compact curves. Their extension theorem for more general degenerations still supplies effective divisors, but leaves open whether open Riemann surfaces can occur as components. This distinction is central here. We construct divisors directly on the infinite cyclic cover of the given surface, and prove that every component is compact. The estimates must work for all the compact Chern classes needed in the eventual curve count.

The weighted analysis uses the Fourier–Laplace method for periodic ends developed by Taubes [27]; the treatment of elliptic complexes by Mrowka–Ruberman–Saveliev [24] provides a related framework. We prove the Dolbeault statements needed here from acyclicity of compact character bundles, including the continuation through the trivial character. For compactness of curve components, the argument combines residue currents, analytic-cycle compactness [4, 17], and Fourier estimates on source cylinders. A periodic gauge estimate uses exponential integrability [31] to control the remaining plane case.

The argument

After Enoki's reduction we assume that no nonzero effective divisor has square zero. Let $Y \rightarrow X$ be the infinite cyclic cover associated with a primitive integral class in $H^1(X; \mathbb{Z})$, and let T generate its deck group. We choose a proper height h satisfying $h \circ T = h + 1$, with positive constant complex Laplacian for a lifted Gauduchon metric. The positive and negative ends mean $h \rightarrow +\infty$ and $h \rightarrow -\infty$, respectively. Write K_Y for the canonical bundle and also for its first Chern class in intersection products.

Compact classes and holomorphic sections. A compact line-bundle datum is a smooth line bundle $E \rightarrow Y$ with specified trivializations outside a compact set. Those trivializations define a relative Chern class $e \in H_c^2(Y; \mathbb{Z})$. The finite cyclic quotient $X_n = Y/\langle T^n \rangle$ has a negative orthonormal intersection basis of size $nb_2(X)$. Suitable sums of its basis classes become trivial at a fixed cutting hypersurface. Cutting there gives $d \geq nb_2(X) - C_0$ independent compact classes e_i , with C_0 independent of n , and

$$e_i^2 = K_Y \cdot e_i < 0.$$

Section 3 constructs these data and retains their prescribed end trivializations.

The absence of effective square-zero divisors makes the nontrivial character-twisted Dolbeault complexes on X acyclic. In Section 4, Fourier transformation in the deck variable then gives Fredholm complexes on Y with exponential conditions at both ends. Each compact datum admits a holomorphic structure approaching the trivial operator at any prescribed finite exponential rate. For the classes above, compact excision gives weighted index one. The remaining ingredients are degree-two vanishing and uniqueness from the positive-end constant coefficient: together they force a one-dimensional section space. Normalize its section s to approach one at the positive end. Its negative-end constant must be zero; otherwise its divisor would be compact, have relative class e , and project to an effective divisor of negative canonical degree on X . Thus the zero divisor $D = (s = 0)$ is bounded above in height.

Differentials obstruct open components. The components of D are not yet known to be compact. First, Section 5 bounds the area of D on the height slabs $[-\ell - 1, -\ell]$, $\ell \geq 0$, by $C_{De} A^\ell$, with A depending only on the fixed geometry. The holomorphic correction rate can therefore be chosen sufficiently large after A is fixed. If $f : R \rightarrow B \subset Y$ is the proper normalization of a component, a holomorphic differential on R defines a $(2, 1)$ -current on Y by integration against pulled-back test forms. Exponential mass bounds place these currents in one fixed weighted cohomology space.

There are two different cohomological obstructions. For scalar currents the degree-one cohomology group is zero. An exact residue current has a primitive vanishing on the positive end; analytic continuation forces that primitive to be supported on the curve, where a local distribution calculation rules it out. In particular there are no nonzero L^2 holomorphic differentials on R , so R has genus zero. A compact normalization is therefore $\mathbb{P}^1$, and an open one is a domain in $\mathbb{P}^1$.

Two uses of the cylinder estimate. Section 6 proves a bound for a differential's mass over a target height slab in terms of the curve's area over a fixed enlarged slab. On the logarithmic source cylinder, with coordinate $v = t + i\theta$ and θ taken modulo 2π , it allows a multiplier with uniformly bounded L^2 norm on each unit t -strip. Its first application uses multiplier one: if the domain $R \subset \mathbb{P}^1$ omits two points, put them at 0 and ∞ and use dz/z on the resulting domain in $\mathbb{C}^*$. The estimate gives an exponentially bounded scalar residue current, contradicting the preceding obstruction. An open normalization must consequently be the plane.

For a plane component B , consider translates $B_k = T^k B$ that are not contained in D , with normalization maps $f_k : \mathbb{C} \rightarrow B_k$. The bundle-valued differentials $(f_k^* s) dz$ are nonzero. Properness and the upper height bound send $|z| \rightarrow \infty$ to the negative end, where $s \rightarrow 0$. A periodic gauge converts this limit into a holomorphic function vanishing to order at least one at infinity, canceling the growth of dz in the coordinate $v = \log z$. The same cylinder estimate, now with an L^2 -bounded multiplier supplied by the gauge, bounds all these currents with the same exponent. Their coefficients lie in one fixed bundle E . Its degree-one current cohomology is finite dimensional, and the residue argument makes currents on distinct curves linearly independent. Infinitely many translates give the contradiction.

The uniform cylinder bound is what permits both applications. Its proof separates source strips having a fixed energy cost from long chains whose images lie in small coordinate charts. Along each such chain, the derivative has zero constant angular Fourier mode. The remaining modes give a summable derivative bound independent of the chain length, so the change of target height along the chain is uniformly bounded. No bound on the moduli of thin necks is required.

Counting curves downstairs. Every component of every D_i is now compact rational, although a whole divisor can still have infinitely many components. In Section 7, their intersections with compact test classes are organized over $\mathbb{Q}[t, t^{-1}]$, where t records translation by T^n . The pairing of the classes e_i becomes nonsingular over $\mathbb{Q}(t)$. If there were $r < b_2(X)$ rational curves downstairs, their lifts would give at most nr translation orbits, too few to support $d \geq nb_2(X) - C_0$ independent classes for large n . The resulting lower bound, together with the curve-class upper bound when no effective divisor has square zero, gives exactly $b_2(X)$ rational curves. Dloussky–Oeljeklaus–Toma then supplies the shell.

Organization and conventions

Section 2 establishes the compact-surface facts, the height, and the finite cyclic quotients. Section 3 constructs the compact Chern classes, and Section 4 produces their holomorphic sections. Section 5 develops the growth and residue obstructions. Section 6 proves the cylinder estimate and compactness of every divisor component. Section 7 counts the curves, proves the main theorem, and derives the two introductory consequences.

All intersection products are taken with the complex orientation. Integral cohomology classes used in diagonal bases are understood modulo torsion; prescribed end trivializations retain the compact support information on the cover. The complex Laplacian has the subharmonic sign. Unless stated otherwise, geometric norms on the cover come from a fixed Hermitian metric on the compact base.

2 Surface invariants and the cyclic cover

Throughout, a curve means a reduced irreducible compact analytic curve unless another meaning is specified. A rational curve is a curve whose normalization is $\mathbb{P}^1$; the curve itself is allowed to be singular. We orient every complex surface by its complex structure, and write a dot for the intersection pairing. We use K_X both for the canonical line bundle and for its first Chern class when a cohomological operation makes the meaning unambiguous.

The compact-surface identities of Kodaira give, under the hypotheses of Theorem 1.1,

$$h^{0,1}(X) = 1, \quad h^{1,0}(X) = h^{2,0}(X) = 0, \quad b_2^+(X) = 0, \quad \chi(\mathcal{O}_X) = 0, \quad K_X^2 = -b_2(X). \quad (2.1)$$

These are the numerical identities for class VII surfaces [21, Theorem 3]; see also [3]. In particular the intersection form on $H^2(X; \mathbb{R})$ is negative definite. Our two surface-theoretic reductions are the following established results.

Proposition 2.1 (Surface reductions). *Let V be a minimal compact complex surface with $b_1(V) = 1$, $\kappa(V) = -\infty$, and $b_2(V) > 0$.*

- (i) *If V has a nonzero effective divisor C with $C^2 = 0$, then V has a global spherical shell.*
- (ii) *If V has exactly $b_2(V)$ rational curves, then V has a global spherical shell.*

Here the conclusion supplies an embedded open neighborhood of $S^3 \subset \mathbb{C}^2 \setminus \{0\}$ with connected complement in V .

The first assertion is the square-zero-divisor case of Enoki's theorem [14], stated in this form in [30, p. 1758]. The second is the Main Theorem of Dloussky–Oeljeklaus–Toma [11, p. 283]; their definition on p. 284 includes the connected-complement condition. These statements impose no condition on the fundamental group beyond the displayed surface hypotheses. The original surface X satisfies all of them. We may therefore assume henceforth that

$$C^2 < 0 \quad \text{for every nonzero effective divisor } C \text{ on } X. \quad (2.2)$$

Indeed negative definiteness leaves only the already settled square-zero case as an alternative.

Lemma 2.2 (Canonical basis and curve degrees). *Put $b = b_2(X)$. There is a basis $u_1, \dots, u_b$ of $H^2(X; \mathbb{Z})/\text{Tors}$ with*

$$u_i \cdot u_j = -\delta_{ij}, \quad K_X = u_1 + \dots + u_b \pmod{\text{Tors}}. \quad (2.3)$$

For every curve $B \subset X$,

$$B^2 < 0, \quad (K_X + B) \cdot B \leq 0, \quad K_X \cdot B \geq 0. \quad (2.4)$$

Consequently every effective divisor has nonnegative canonical degree.

Proof. Donaldson's diagonalization theorem in its arbitrary-fundamental-group version applies to the smooth, closed, oriented four-manifold X , whose intersection form is negative definite by (2.1). Precisely, we use Theorem 1 of [13], on integral cohomology modulo torsion. Its surface application is also recorded in [30, §1.1, p. 1756]. Choose a negative orthonormal integral basis and write $K_X = \sum_i k_i u_i$ modulo torsion. The canonical class is characteristic: its mod-two reduction is the second Stiefel–Whitney class. Thus every k_i is odd. Since $K_X^2 = -b$, we have $\sum_i k_i^2 = b$, so $k_i = \pm 1$ for every i . Replacing each basis vector by its negative when necessary gives (2.3).

If $[B] = \sum_i a_i u_i$ modulo torsion, the integers a_i give

$$(K_X + B) \cdot B = -\sum_i a_i(a_i + 1) \leq 0.$$

The strict inequality $B^2 < 0$ follows from (2.2). The adjunction formula for a possibly singular integral curve is

$$K_X \cdot B = 2p_a(B) - 2 - B^2.$$

If $p_a(B) \geq 1$, this is positive. If $p_a(B) = 0$, the normalization-genus formula $p_a(B) = g(\tilde{B}) + \sum_x \delta_x$ shows that B is a smooth rational curve. Minimality and $B^2 < 0$ then give $B^2 \leq -2$, so its canonical degree is again nonnegative. Additivity proves the last assertion. The adjunction and characteristic-class formulas used here are the standard compact-surface formulas, as in [3]. $\square$

Since $H^1(X; \mathbb{Z}) \simeq \mathbb{Z}$, choose a primitive generator α . The induced epimorphism $\pi_1(X) \rightarrow \mathbb{Z}$ defines a connected regular covering

$$\pi : Y \longrightarrow X, \quad \text{Deck}(Y/X) = \langle T \rangle \simeq \mathbb{Z}.$$

Only this quotient of the fundamental group is used. Choose a smooth real closed representative of α and integrate its pullback to obtain a smooth function $h_0 : Y \rightarrow \mathbb{R}$ satisfying $h_0(Ty) = h_0(y) + 1$. Later we may reverse both the generator and the sign of the height. The Hermitian metric on the underlying surface Y will always be lifted from X ; auxiliary bundle metrics will be specified separately.

For $\lambda \in \mathbb{C}^*$, let P_λ be the flat holomorphic line bundle whose sections pull back to holomorphic functions with $f(Ty) = \lambda f(y)$. This convention fixes the sign of the character. These bundles are smoothly trivial: if $e^u = \lambda$, the function e^{uh_0} is a nowhere-zero smooth equivariant section. In particular $c_1(P_\lambda) = 0$ integrally.

Lemma 2.3 (Character cohomology). *For $\lambda \neq 1$ and every $q = 0, 1, 2$,*

$$H^q(X, P_\lambda) = 0, \quad H^q(X, K_X \otimes P_\lambda) = 0. \quad (2.5)$$

For every $\lambda \in \mathbb{C}^$, including $\lambda = 1$,*

$$H^0(X, K_X \otimes P_\lambda) = 0. \quad (2.6)$$

At the trivial character, $H^0(X, \mathcal{O}_X) = \mathbb{C}$, $H^1(X, \mathcal{O}_X) = \mathbb{C}$, and $H^2(X, \mathcal{O}_X) = 0$. If a holomorphic function f on Y satisfies $f \circ T - f = c$ for a constant c , then f is constant and $c = 0$. More generally, a holomorphic function annihilated by a positive power of $T^ - 1$ is constant.*

Proof. A nonzero section of P_λ has an effective zero divisor of Chern class zero. A nonempty such divisor contradicts (2.2); hence the section is nowhere zero. Its lifted function f has a logarithmic differential df/f , defined without a choice of logarithm. Equivariance makes df/f invariant under T , so it descends to a holomorphic one-form on X . By (2.1) this form vanishes. Thus f is a nonzero constant and $\lambda = 1$. This proves the degree-zero assertion for P_λ .

A nonzero section of $K_X \otimes P_\lambda$ would have zero divisor C with $[C] = K_X$ in real cohomology. If C is nonempty, then $K_X \cdot C = K_X^2 = -b < 0$, contradicting Lemma 2.2. If C is empty, the bundle is holomorphically trivial and its real first Chern class vanishes, which also contradicts $K_X^2 < 0$. This proves (2.6) for all characters.

Serre duality gives $H^2(X, P_\lambda) \simeq H^0(X, K_X \otimes P_{\lambda^{-1}})^* = 0$. Riemann–Roch for line bundles gives $\chi(P_\lambda) = \chi(\mathcal{O}_X) = 0$, since $c_1(P_\lambda) = 0$. Thus $H^1(X, P_\lambda) = 0$ if $\lambda \neq 1$. Serre duality once more gives all the canonical-bundle vanishings in (2.5). The assertions at $\lambda = 1$ follow from (2.1) and connectedness of X .

Finally, $f \circ T - f = c$ makes df invariant, hence the pullback of a holomorphic one-form on X . It is zero, so f is constant. For the more general assertion, write $\Delta = T^* - 1$. If a least exponent $m > 1$ had $\Delta^m f = 0$ and $\Delta^{m-1} f \neq 0$, the last function would descend to a nonzero constant on X . The function $\Delta^{m-2} f$ would then have a nonzero constant difference, which was just excluded. Thus $m = 1$, and f descends to a constant on X . $\square$

Lemma 2.4 (A height with positive complex Laplacian). *There are a Gauduchon metric on X , a choice of T , and a smooth proper height $h : Y \rightarrow \mathbb{R}$ such that*

$$h(Ty) = h(y) + 1, \quad Lh = c > 0. \quad (2.7)$$

Here L is the complex Laplacian with the subharmonic sign. All positive order derivatives of h are uniformly bounded in the lifted metric.

Proof. Gauduchon's theorem supplies a Hermitian fundamental form ω with $\partial\bar{\partial}\omega = 0$ [15]. Fix the convention $d^c = i(\bar{\partial} - \partial)$, so $dd^c = 2i\partial\bar{\partial}$, and put $L = \Lambda_\omega dd^c$. This is elliptic and has the subharmonic sign. Its kernel on real functions on the compact connected surface consists of constants by the maximum principle. Its index is zero: its principal symbol is that of a positive multiple of the scalar Laplacian, and a homotopy of its lower-order terms to that Laplacian preserves the elliptic index. Moreover,

$$\int_X Lu \frac{\omega^2}{2} = \int_X dd^c u \wedge \omega = \int_X u dd^c \omega = 0.$$

The cokernel therefore consists exactly of constants. These are also the Gauduchon-Laplacian solvability facts stated in [6, p. 289].

The one-forms dh_0 and $d^c h_0$, and the function Lh_0 , are periodic. Let c be the mean of Lh_0 on X . The preceding solvability yields a smooth function v on X such that $L(v \circ \pi) = c - Lh_0$. Set $h = h_0 + v \circ \pi$. If $c = 0$, the descended real $(1, 1)$ -form $\eta = dd^c h$ would be both exact and primitive. On a Hermitian surface a real primitive $(1, 1)$ -form is anti-self-dual, and consequently

$$0 = \int_X \eta \wedge \eta = - \int_X |\eta|^2 \frac{\omega^2}{2}.$$

The first equality follows from exactness and Stokes' theorem. Thus $dd^c h = 0$, so ∂h is an invariant holomorphic one-form. Its descended form is nonzero: otherwise the real function h would be constant, contrary to its unit increment under T . This contradicts $h^{1,0}(X) = 0$. Hence $c \neq 0$; replacing (T, h) by $(T^{-1}, -h)$ if necessary makes $c > 0$.

Every positive order derivative of h is periodic and therefore bounded. To see properness, choose a compact set $F \subset Y$ whose deck translates cover Y , possible because X is compact. The height is bounded on F . For any bounded interval I , only finitely many translates $T^j F$ can meet $h^{-1}(I)$, by the unit-increment identity. The closed set $h^{-1}(I)$ is therefore compact. This proves properness. $\square$

Lemma 2.5 (Finite cyclic quotients). *For every $n \geq 1$, put $S = T^n$ and $X_n = Y/\langle S \rangle$, and let $p_n : X_n \rightarrow X$ be the degree- n covering. Then X_n is a minimal class VII surface and*

$$\chi(\mathcal{O}_{X_n}) = 0, \quad b_1(X_n) = 1, \quad b_2(X_n) = nb_2(X), \quad K_{X_n} = p_n^* K_X.$$

In particular its integral intersection lattice modulo torsion has a negative orthonormal basis whose sum is K_{X_n} modulo torsion.

Proof. The unramified holomorphic covering is a local biholomorphism, so its tangent and canonical bundles are pullbacks. Compact-complex Riemann–Roch, or the degree formula for the Todd class [2, §4], gives $\chi(\mathcal{O}_{X_n}) = n\chi(\mathcal{O}_X) = 0$. If σ were a nonzero section of $K_{X_n}^{\otimes m}$ for $m > 0$, the product of its n deck transforms would be a nonzero invariant section of $K_{X_n}^{\otimes mn}$. It would descend to a nonzero section of $K_X^{\otimes mn}$, contradicting $\kappa(X) = -\infty$. Thus $\kappa(X_n) = -\infty$ and $p_g(X_n) = 0$; the Euler characteristic then gives $q(X_n) = 1$.

The surface X_n is non-Kähler: averaging a hypothetical Kähler form over its finite deck group would give an invariant Kähler form and hence a Kähler form on X . That is impossible because $b_1(X) = 1$. The compact-surface degree-one Hodge identities give $b_1 = 2q$ for a surface with even b_1 and $b_1 = 2q - 1$ for a surface with odd b_1 [3]. By the even-first-Betti Kähler criterion [6, Theorem 11], $b_1(X_n)$ is odd. Hence $b_1(X_n) = 2q(X_n) - 1 = 1$. The topological Euler characteristic of a closed oriented four-manifold is $2 - 2b_1 + b_2$. It multiplies by the degree of a finite covering, so $b_2(X_n) = nb_2(X)$.

For minimality, suppose that $C \subset X_n$ were a smooth rational curve with $C^2 = -1$. Adjunction would give $K_{X_n} \cdot C = -1$. Its image is a curve $B \subset X$, and if the degree of $C \rightarrow B$ is $d > 0$, the projection formula and Lemma 2.2 give

$$-1 = K_{X_n} \cdot C = p_n^* K_X \cdot C = d(K_X \cdot B) \geq 0,$$

a contradiction. Thus X_n is minimal. Its class VII numerical identities and the argument of Lemma 2.2 now give the asserted canonical basis. No claim about the integral homology of Y enters this argument. $\square$

For reference, the notation used on the cover is summarized here.

$X, b = b_2(X)$	original minimal class VII surface and its second Betti number
$Y \xrightarrow{\pi} X, T$	infinite cyclic cover and unit-height deck generator
h, K_Y	proper height and canonical bundle on Y
$X_n = Y/\langle T^n \rangle$	finite cyclic quotient of degree n over X
P_λ	flat holomorphic character line bundle on X

3 Compact Chern classes

We first construct many compactly supported line-bundle data on Y whose Chern classes satisfy $e^2 = K_Y \cdot e < 0$. This equality will make the weighted index in the next section equal to one. Sums of distinct canonical basis elements retain the equality, whereas arbitrary multiples do not. We therefore seek such sums that become trivial along a fixed seam, so that they can be cut open and extended to Y . Here a datum is a smooth complex line bundle E together with a fixed nowhere-zero frame outside a compact set, or, equivalently, fixed frames on both sufficiently distant ends. It has a relative first Chern class

$$e = c_1(E, \text{end frames}) \in H_c^2(Y; \mathbb{Z}).$$

Its image under the forgetful map $j : H_c^2(Y; \mathbb{Z}) \rightarrow H^2(Y; \mathbb{Z})$ is $c_1(E)$. For compactly supported degree-two classes a, b we use the pairing

$$\langle a, b \rangle = \int_Y a \smile j(b), \quad K_Y \cdot a = \int_Y c_1(K_Y) \smile a.$$

Thus every integral here has a compactly supported factor. In de Rham notation these integrals are wedge products of closed representatives; their values are independent of those representatives.

Choose a regular value c of h and put $Z = h^{-1}(c)$. The height is proper by Lemma 2.4, so Z is a compact two-sided oriented three-manifold, possibly disconnected. Its image in each X_n is embedded and will again be denoted by Z . The quotient map identifies the cut-open X_n with

$$W_n = h^{-1}([c, c + n]),$$

with its two boundary copies of Z identified by $S = T^n$. Indeed each S -orbit has one representative with $c < h < c + n$, except for the seam orbits. No connectedness assertion about Z or W_n is needed.

Lemma 3.1 (Many compact classes). *There is a constant $C_0 \geq 0$, independent of n , such that for each n one can choose smooth line-bundle data $E_1, \dots, E_d$ on Y , supported in the interior of W_n , with*

$$d \geq nb_2(X) - C_0. \tag{3.1}$$

They are obtained by cutting line bundles on X_n that are trivialized near Z . Their transferred Chern classes $w_1, \dots, w_d \in H^2(X_n; \mathbb{Z})$ are linearly independent over $\mathbb{Q}$ modulo torsion. Their compact Chern classes e_i satisfy

$$e_i^2 = K_Y \cdot e_i = -m_i < 0 \quad (3.2)$$

for positive integers m_i . Pairings of these compact classes equal the corresponding pairings of the transferred classes on X_n .

Proof. Write $N = nb_2(X)$. By Lemma 2.5, choose actual integral lifts $u_1, \dots, u_N \in H^2(X_n; \mathbb{Z})$ of a canonical negative orthonormal basis modulo torsion. We first show that every restriction $u_j|_Z$ is torsion.

Let $i : Z \hookrightarrow X_n$ be the inclusion. For $a \in H_2(Z; \mathbb{Q})$, a small displacement in the normal direction gives a representative of i_*a disjoint from a representative on Z . This displacement is isotopic to the inclusion, so $(i_*a)^2 = 0$. Negative definiteness of X_n forces $i_*a = 0$. In particular the whole image of $H_2(Z; \mathbb{Q})$ is zero. By the cohomology–homology pairing over $\mathbb{Q}$, the restriction

$$H^2(X_n; \mathbb{Q}) \longrightarrow H^2(Z; \mathbb{Q})$$

is zero. All integral degree-two classes therefore restrict to the fixed finite group $G = \text{Tors } H^2(Z; \mathbb{Z})$.

Group the chosen basis lifts by their restriction $\tau \in G$. Let $k_\tau \geq 1$ be the order of τ , with order one for the zero element, and let N_τ be the size of its group. Any sum of k_τ distinct basis lifts in this group restricts to zero on Z . If $N_\tau > k_\tau$, these sums span the full rational coordinate space of the group. To check this, take any distinct indices i, j in the group and choose a set J of $k_\tau - 1$ other indices. The difference

$$\left(u_i + \sum_{a \in J} u_a \right) - \left(u_j + \sum_{a \in J} u_a \right) = u_i - u_j$$

lies in the span. Such differences span the coordinate-sum-zero hyperplane, while any one of the indicated sums has nonzero coordinate sum k_τ . If $N_\tau = k_\tau$, there is one such sum and the loss of dimension is $k_\tau - 1$. If $N_\tau < k_\tau$, taking no sum loses at most $k_\tau - 1$ dimensions. Consequently the collection of all these sums spans a space of dimension at least

$$N - C_0, \quad C_0 = \sum_{\tau \in G} (k_\tau - 1).$$

Choose a rationally independent subcollection of the actual sums, and call its elements $w_1, \dots, w_d$. The constant C_0 depends only on the fixed seam Z .

A smooth complex line bundle is classified by its integral first Chern class. Thus w_i is the Chern class of a smooth line bundle L_i on X_n . Since $w_i|_Z = 0$ integrally, L_i is trivial on a collar of Z ; choose one frame there. Cut along Z , retaining the restrictions of this same frame on the two boundary collars of W_n . Lift the cut-open bundle to $W_n \subset Y$ and extend it over the two remaining ends using trivial bundles and these frames. This produces E_i with the stipulated end framings. Its relative class e_i is supported away from the boundary of W_n .

For the pairing assertion, choose a smooth unitary connection on L_i that is trivial in its chosen collar frame. Its normalized curvature is a closed representative of w_i , vanishing near Z . The same form, lifted to the cut-open period and extended by zero, represents e_i . Integrating a wedge product of two such forms on Y is exactly integrating their original wedge product on X_n . The same argument works with the canonical class, whose curvature form pulls back under the covering. If w_i is the sum of m_i distinct canonical basis elements, then

$$w_i^2 = -m_i, \quad K_{X_n} \cdot w_i = -m_i.$$

Torsion parts contribute zero to these integer pairings. This proves (3.2) and the lemma. $\square$

4 Weighted Dolbeault theory on the cyclic cover

Lemma 3.1 supplies smooth end-framed line bundles whose compact Chern classes satisfy $e^2 = K_Y \cdot e < 0$. We now give each such bundle a holomorphic structure and a holomorphic section approaching 1 at the positive end and 0 at the negative end. The structure will approach the trivial one at any prescribed finite exponential rate; that rate remains free throughout this section.

For a general compact datum we prove top-degree cohomology vanishing and show that a holomorphic section is determined by its constant coefficient at the positive end. Compact excision computes its weighted Dolbeault index as $1 + \frac{1}{2}(e^2 - K_Y \cdot e)$. For the classes above this is one, so the vanishing and uniqueness statements give a unique section after normalizing its positive coefficient to 1. The negative canonical degree of e then forces its negative coefficient to vanish.

The periodic input is Lemma 2.3:

$$H^q(X, P_\lambda) = H^q(X, K_X \otimes P_\lambda) = 0 \quad (\lambda \neq 1, 0 \leq q \leq 2), \quad H^0(X, K_X \otimes P_\lambda) = 0 \quad (\lambda \in \mathbb{C}^*). \quad (4.1)$$

The height furnished by Lemma 2.4 satisfies $h \circ T = h + 1$, $Lh = c > 0$, and $|dh| \leq C$. All its positive-order derivatives are bounded. We fix the lifted Hermitian metric and its volume form. The analysis of the ends uses these facts and the absence of additive holomorphic modes established in Lemma 2.3.

4.1 Spaces, adjoints, and Fredholm complexes

For a compact line-bundle datum E as defined in Section 3, choose a Hermitian metric and a smooth connection agreeing with the standard ones in its prescribed end frames. Changing these choices on a compact set gives equivalent Sobolev norms. For $s \in \mathbb{Z}$, let W^s denote the Sobolev space defined by a uniformly locally finite lifted coordinate cover and partition of unity. For negative s use distributional duals of the positive-order local norms. The lifted geometry has bounded derivatives, positive injectivity radius, and uniformly finite overlap, so the usual localization, multiplication, and elliptic estimates have constants independent of the lifted chart.

For real numbers a, b , choose a smooth function $\phi_{a,b}$ equal to ah on a sufficiently far positive tail and to $-bh = b|h|$ on a sufficiently far negative tail. Its positive-order derivatives are bounded. Set

$$W_{a,b}^s(\Lambda^{p,q} \otimes E) = \{u : e^{-\phi_{a,b}} u \in W^s(\Lambda^{p,q} \otimes E)\}, \quad \|u\|_{s;a,b} = \|e^{-\phi_{a,b}} u\|_{W^s}. \quad (4.2)$$

Any change in $\phi_{a,b}$ on a compact set gives the same space with an equivalent norm. Positive a or b permits growth at the corresponding end, and negative a or b requires decay. A single linear weight $e^{-\rho h}$ has $(a, b) = (\rho, -\rho)$. We call a pair admissible when $a \neq 0$ and $b \neq 0$.

Suppose for the moment that E has an integrable operator d_E of type $(0, 1)$ which, in its end trivializations, is

$$d_E = \bar{\partial} + A_E \wedge, \quad |\nabla^j A_E| \leq C_j e^{-\gamma|h|} \quad (j \geq 0) \quad (4.3)$$

for some $\gamma > 0$. The same discussion applies to E^{-1} , to the trivial bundle, and after tensoring by K_Y . For a fixed integer s , the stepped Sobolev complex is the bounded complex

$$\mathcal{C}_{a,b,s}^{p,q}(E) = W_{a,b}^{s-q}(\Lambda^{p,q} \otimes E), \quad d_E : \mathcal{C}_{a,b,s}^{p,q}(E) \longrightarrow \mathcal{C}_{a,b,s}^{p,q+1}(E), \quad 0 \leq q \leq 2. \quad (4.4)$$

The decrease of Sobolev order with q makes the differential bounded; it will also allow the same theory to handle residue currents. The cohomology will be denoted by $H_{a,b}^{p,q}(Y, E)$; its independence of s follows below. An alternative realization uses the maximal closed operators d_E in the equal-order weighted L^2 spaces. We shall verify that this Hilbert complex has the same cohomology.

The analytic background used here is the compact elliptic Hodge theorem, the local elliptic estimate (including its distributional Sobolev versions), Rellich's compact embedding on relatively compact sets, and Plancherel's theorem. On a compact manifold, the Hodge theorem for an elliptic complex provides a finite-dimensional smooth harmonic space and a Green operator of order -2 , with $I = P + dd^*G + d^*Gd$. The local elliptic estimate raises regularity by the order of an elliptic operator when its right hand side has the corresponding regularity. These results apply to smooth bundles and smooth elliptic operators, with no Kähler hypothesis; for compact Hermitian Dolbeault theory see [7, Chapter VI, Theorems (7.1)–(7.2)]. We give the periodic and two-ended arguments, rather than import a theorem about a different complex. The periodic-end Fredholm method goes back to Taubes [27, Lemma 4.3]; the de Rham formulation in [24, Proposition 2.1] is useful background. The two-ended Dolbeault argument needed here is proved below.

Proposition 4.1. *For $p = 0$ or $p = 2$, every admissible pair (a, b) and every $s \in \mathbb{Z}$, the complex (4.4) has closed ranges and finite-dimensional cohomology. Its cohomology agrees with that of the maximal weighted L^2 complex and is represented by forms which belong to $W_{a,b}^k$ for every k . There are bounded homotopies of degree -1 which gain one derivative, modulo projections onto these harmonic spaces. For the periodic trivial datum the single-linear-weight complex with $\rho \neq 0$ is acyclic, in every Sobolev realization. The operator bounds may depend on the fixed rates and the datum.*

Proof. First take a periodic complex on the whole cover. Conjugation by $e^{-\rho h}$ changes its differential to

$$d_\rho = d + \rho \bar{\partial} h \wedge.$$

Fourier transformation of the deck variable decomposes this complex as a direct integral of the compact complexes on X with multipliers $e^{\rho + i\theta}$, $0 \leq \theta \leq 2\pi$. One can see both the identification and the norm assertion on a compact fundamental region: the sequence of its translates is an ℓ^2 sequence with values in local Sobolev spaces, and multiplication by $e^{-\rho h}$ changes the norm only by bounded factors on that region. Plancherel's theorem gives the direct integral. The endpoint gauges at $\theta = 0, 2\pi$ agree by the unitary gauge $e^{2\pi i h}$; thus a global logarithm of the multiplier is unnecessary.

By (4.1), every compact complex on this circle is acyclic. Its Hodge operator $d_\rho + d_\rho^*$ is consequently invertible. The compact elliptic estimate, the bounded inverse theorem, and a finite covering of the parameter circle give a uniform inverse from L^2 to W^1 . For completeness, locally in the parameter an already invertible operator remains invertible by a Neumann series; the difference of two nearby operators is a small bounded map $W^1 \rightarrow L^2$. Applying the compact elliptic estimate to the inverse gives the uniform higher-order bounds. Fourier inversion therefore gives an inverse for the conjugated periodic Hodge operator, and the compact homotopy identities integrate to an acyclic complex with a homotopy gaining one derivative. Duality gives the same bounds at negative Sobolev orders.

The adjoint used for a general weight is specifically the adjoint in that weighted norm:

$$d_{E,\phi}^* = e^{2\phi} d_E^* e^{-2\phi}. \quad (4.5)$$

Thus $d_E + d_{E,\phi}^*$, conjugated by $e^{-\phi}$, is $\tilde{d}_E + \tilde{d}_E^*$, where $\tilde{d}_E = e^{-\phi} d_E e^\phi$. At the positive end its periodic limit is the operator just considered with $\rho = a$; at the negative end it is the one with $\rho = -b$. The difference has order zero and all its coefficients tend to zero, by (4.3).

Here is the parametrix argument explicitly. On each tail use the inverse of the corresponding whole-cover periodic limit, with nested cutoffs equal to one sufficiently far along that tail. On a relatively compact middle region use an interior elliptic parametrix. Compose their sum with the

conjugated Hodge operator on either side. Commutators with the cutoffs are order-zero operators supported in compact collars. Following or preceding the order- -1 parametrices they are compact, by Rellich's theorem. Multiplication by a coefficient tending to zero is compact from W^1 to L^2 : truncate it to a compact set, apply Rellich there, and bound the remaining operator norm by its supremum on the tails. This treats the end errors as well. The resulting left and right parametrices are inverses modulo compact operators. In particular the Hodge operator is Fredholm from W^1 to L^2 .

This construction also proves the useful estimate

$$\|u\|_{1;a,b} \leq C(\|(d_E + d_{E,\phi}^*)u\|_{0;a,b} + \|u\|_{L^2(C)}) \quad (4.6)$$

for a fixed compact set C , and gives the analogous estimate for the adjoint. The global first-order elliptic estimate shows that the maximal domain of the conjugated Hodge operator is W^1 . To justify the assertion for an initially distributional element of that domain, apply the local estimate on the uniform charts and sum; cutoffs in h/R then exhaust Y , with bounded commutators tending to zero. The same cutoffs show that compactly supported smooth sections are a core. The operator is therefore self-adjoint on W^1 .

Because $d_E^2 = 0$, its square is the degree-preserving Hodge Laplacian. Fredholmness and self-adjointness give a spectral gap away from its finite-dimensional kernel. Let P be the orthogonal projection onto that kernel and let G be the inverse of the Laplacian on its orthogonal complement, zero on the kernel. Uniform elliptic estimates give

$$G : W_{a,b}^t \longrightarrow W_{a,b}^{t+2} \quad (t \in \mathbb{Z}), \quad Q = d_{E,\phi}^* G : W_{a,b}^t \longrightarrow W_{a,b}^{t+1}.$$

For negative t , these follow also by transposing the positive-order estimates after conjugation. Elements of $\text{im } P$ have all weighted Sobolev orders, so P extends to all these spaces. Commutation with the differential, first on smooth vectors and then by density, gives

$$I = P + d_E Q + Q d_E. \quad (4.7)$$

On the stepped complex Q maps degree q into degree $q - 1$ with exactly the required gain of one derivative. A closed element has zero class precisely when its projection under P is zero. This proves closed ranges, finite-dimensionality, and independence of the Sobolev realization. It proves the same statements for the maximal L^2 complex, completing the proof. $\square$

Lemma 4.2 (Weighted duality). *For $p \in \{0, 2\}$ and $0 \leq q \leq 2$, the integration pairing gives a perfect pairing*

$$H_{a,b}^{p,q}(Y, E) \times H_{-a,-b}^{2-p,2-q}(Y, E^{-1}) \longrightarrow \mathbb{C}. \quad (4.8)$$

Here both pairs of rates are admissible. The statement concerns the cohomology classes in any of the stepped realizations above; the pairing is evaluated on their smooth weighted harmonic representatives.

Proof. Choose $\phi_{-a,-b} = -\phi_{a,b}$, so their exponential weight factors are reciprocal. On weighted L^2 representatives the integrand is absolutely integrable by Cauchy–Schwarz, since the two weights cancel. Integration by parts is valid by inserting the cutoffs in h/R : their derivatives are bounded and the L^2 tails tend to zero. Thus exact classes pair trivially with closed classes. The Hermitian Hodge star and the bundle metric give an antilinear map J from (p, q) forms with values in E to complementary forms with values in E^{-1} , characterized by the pointwise Hermitian inner product under wedge integration. The map $u \mapsto e^{-2\phi} J u$ takes the weighted harmonic equations to the opposite-weight harmonic equations; this follows immediately by transposing d_E and using (4.5).

It is an isomorphism, and the pairing of u with its image is its weighted squared norm, up to the harmless fixed degree sign. Harmonic representatives therefore give a nondegenerate pairing. Both spaces are finite-dimensional by Proposition 4.1. At the cochain level, the dual of the order $s - q$ term has order $q - s$. Writing the complementary degree as $2 - q$ shows that the dual stepped complex has base order $2 - s$, since $(2 - s) - (2 - q) = q - s$. Thus opposite weights also have this complementary Sobolev shift; independence of the shift, already proved above, is what allows the notation in (4.8) to suppress it. $\square$

4.2 Continuation in the multiplier and positive-tail uniqueness

Fredholmness gives finite-dimensional kernels, but the index argument requires more: scalar-type sections must have a single constant coefficient at the positive end, and canonical sections must vanish there. We obtain these conclusions by improving exponential rates. First, the sign $Lh > 0$ converts decay at every rate into vanishing on the positive tail. We will then prove the required decay by continuing the Fourier–Laurent transform in its multiplier.

Lemma 4.3 (Positive-tail nondecay). *Let V be a holomorphic line bundle on $\{h > H_0\}$, with a smooth Hermitian metric of bounded Chern curvature. If a holomorphic section s satisfies, for every $N > 0$,*

$$|s(x)| \leq C_N e^{-Nh(x)} \quad (h(x) \text{ sufficiently large}),$$

then s vanishes on a positive tail. On every connected region on which it is holomorphic and which meets that tail it consequently vanishes identically.

Proof. The logarithm of the norm of a holomorphic section satisfies $L \log |s| \geq -M$ in distributions, including at its zeros, for a fixed M depending on the curvature bound. In a holomorphic frame this is the fact that the logarithm of the modulus of a holomorphic function is plurisubharmonic, plus the bounded curvature contribution of the metric. Consequently $u = \log |s| + (M/c)h$ is subharmonic for L . The chain rule and the bound on dh give, for a sufficiently small $\eta > 0$,

$$L(e^{-\eta h}) = e^{-\eta h} (-\eta c + \eta^2 \sigma_L(dh, dh)) \leq 0,$$

where $\sigma_L(dh, dh)$ is bounded and nonnegative.

Choose a regular value $h_0 > H_0$ and a finite upper bound C for u on its compact level. Given $A > \max(1, -C)$, choose a regular $H > h_0$ so large that $u \leq -A$ on $\{h = H\}$. Such an H exists by the assumed decay. The superharmonic function

$$v_A = -A + (A + C)e^{-\eta(h-h_0)}$$

has value C on the lower boundary and is at least $-A$ on the upper boundary. The maximum principle on the compact slab gives $u \leq v_A$. It applies to the upper-semicontinuous subharmonic logarithm with its value $-\infty$ at zeros, either directly or by local regularization. For a fixed point of height greater than h_0 , take H beyond that point and let $A \rightarrow \infty$. Since $1 - e^{-\eta(h-h_0)} > 0$, we obtain $u = -\infty$ there. No connectedness assumption on the level sets was used. $\square$

Here and below, a section has *finite exponential growth on a tail* if it belongs there to some weighted Sobolev space of finite order and finite real rate. Distributional holomorphic sections are smooth by local elliptic regularity. The uniform local estimates then promote this membership to every Sobolev order on a shorter tail, and to pointwise bounds of the corresponding exponential size.

Lemma 4.4 (Degree-zero continuation). *Let V_0 be either $\mathcal{O}_Y$ or K_Y with its periodic holomorphic operator d . Suppose on one tail that*

$$(d + B\wedge)s = 0, \quad |\nabla^j B| \leq C_j e^{-\gamma|h|}, \quad j \geq 0,$$

where the perturbed operator is integrable and $\gamma > 0$. If s has finite exponential growth, its allowed rate can be improved by any amount strictly less than γ , except that a contour passing the unit multiplier can contribute a constant when $V_0 = \mathcal{O}_Y$. There is no degree-zero contribution for $V_0 = K_Y$.

In the scalar case, after finitely many such improvements there is a constant $c_\pm$ on the relevant end and an exponentially decaying remainder, with decay in all derivatives. If that constant is zero, the section has every finite exponential decay rate on that end. In the canonical case it always has every finite exponential decay rate.

Proof. We transform a cut-off section to the compact base, continue the resulting solutions through the exceptional multiplier, and recover the improved rates by contour inversion.

The transformed equation. Consider first the positive tail and choose a cutoff χ equal to one far along it and zero off a slightly longer tail. In the periodic end frame, extend $u = \chi s$ by zero to Y . Then

$$du = f = (d\chi)s - \chi Bs. \quad (4.9)$$

If s belongs to rate r , multiplication by B puts the second term at rate $r - \gamma$; the first term is compactly supported. Estimates with any strictly intermediate rate avoid boundary-convergence issues.

For a positive-tail-supported form v , its Fourier–Laurent transform is, initially in a right half-plane,

$$\hat{v}(z)(x) = \sum_{k \in \mathbb{Z}} e^{-z(h(x)+k)} (T^k)^* v(x). \quad (4.10)$$

The expression is invariant under T , hence a form on X . If v has rate r , Cauchy–Schwarz on the translates gives normal convergence in compact-manifold Sobolev norms when $\operatorname{Re} z > r$. The differentiated series converges on every smaller half-plane, including any polynomial factors in k , so it is holomorphic there. The transform of (4.9) is

$$d(z)\hat{u}(z) = \hat{f}(z), \quad d(z) = d + z\bar{\partial}h \wedge. \quad (4.11)$$

Its multiplier is $\lambda = e^z$; this also fixes our sign convention. The source is holomorphic on $\operatorname{Re} z > r - \gamma$. Moreover $d(z)\hat{f}(z) = 0$ there: the identity holds in the initial half-plane because $d^2u = 0$, and extends by the identity theorem.

Continuation on the compact base. For $e^z \neq 1$, compact acyclicity gives a unique degree-zero solution of (4.11). It depends holomorphically on z . To verify the latter assertion without choosing a nonholomorphic family of Hodge inverses, fix z_0 where degree-zero cohomology vanishes. Compact ellipticity makes

$$B_0 = (d(z_0)^* d(z_0))^{-1} d(z_0)^*$$

a bounded left inverse, from W^t to W^{t+1} . Writing $d(z) = d(z_0) + (z - z_0)A$, any existing solution obeys

$$\hat{u}(z) = [I + (z - z_0)B_0A]^{-1} B_0 \hat{f}(z). \quad (4.12)$$

The right side is holomorphic near z_0 . In the canonical case degree-zero cohomology also vanishes at $e^{z_0} = 1$, so the same formula continues the solution across that point. The full equation, not only

its image under B_0 , holds by analytic continuation. Higher canonical cohomology at this parameter creates no pole in this degree-zero calculation.

In the scalar case the constants prevent such a left inverse at the exceptional point. A local gauge reduces that point to $z_0 = 0$. We first solve for the component orthogonal to the constants, then determine the constant coefficient. Let P_0 be projection onto the constants and use the compact left inverse B_0 with $B_0 d = I - P_0$. Write a prospective solution as $c + v$, with v orthogonal to the constants, and put

$$R(z) = (I + zB_0A)^{-1}, \quad v = R(z)B_0\hat{f} - zcR(z)B_0A1.$$

The inverses here act on the complement of the constants. Set $w(z) = 1 - zR(z)B_0A1$; the candidate solution is $R(z)B_0\hat{f} + cw(z)$. The class of $A1 = \bar{\partial}h$ in $H^1(X, \mathcal{O}_X)$ is nonzero. Indeed, if $\bar{\partial}h = \bar{\partial}v$ for a function on X , then $h - v$ would be a holomorphic function on Y changing by 1 under T , contrary to Lemma 2.3. Choose a continuous scalar functional ℓ on degree-one forms which annihilates $\text{im } d$ and for which $\ell(A1) \neq 0$; pairing with the harmonic representative of this nonzero class gives such a functional. Applying ℓ to the transformed equation determines the candidate coefficient

$$c(z) = \frac{\ell(\hat{f} - d(z)R(z)B_0\hat{f})}{\ell(d(z)w(z))}, \quad \ell(d(z)w(z)) = z\ell(A1) + O(z^2). \quad (4.13)$$

This coefficient is meromorphic with at most a simple pole. For every sufficiently small nonzero z , compact acyclicity already supplies a unique solution of the full transformed equation. That solution must satisfy both the complement equation and the displayed scalar equation, so it equals our candidate. Thus the candidate solves the full equation on the punctured neighborhood and continues it meromorphically across zero. The residue is a constant, since the pole part in the differential equation is annihilated by d . This calculation supplies the required finite-order meromorphic continuation without applying a Fredholm theorem to the overdetermined map $d : W^{t+1}(\Lambda^{0,0}) \rightarrow W^t(\Lambda^{0,1})$.

Recovering the end expansion. For a desired positive-end rate r_1 with $r - \gamma < r_1 < r$, choose a contour real part $\sigma \neq 0$ such that

$$r - \gamma < \sigma < r_1 < r.$$

Start from any vertical line with real part greater than r . Both lines lie in the source's domain of holomorphy, and the continued solution has bounded Sobolev norm on each multiplier circle. Moving the inversion contour to real part σ gives, on a fixed fundamental region,

$$(T^k)^*u(x) = \frac{1}{2\pi i} \int_{\sigma - i\pi}^{\sigma + i\pi} e^{z(h(x)+k)} \hat{u}(z)(x) dz, \quad (4.14)$$

plus the residues crossed when moving from the initial line to σ . Horizontal boundary terms cancel in the contour rectangle: the transform changes by the gauge $e^{-2\pi i h}$ under $z \mapsto z + 2\pi i$, and the factor in (4.14) changes by its inverse. This justifies the contour deformation. Cauchy's estimate on the new line bounds its integral contribution on the k th translate by $Ce^{\sigma k}$ in every fixed Sobolev norm. After applying the target weight $e^{-r_1 k}$, its squared norms on the positive translates are summable because $r_1 - \sigma > 0$. The constants in (4.13) are the only residues; there are none in the canonical case.

At the negative end use a cutoff supported in $\{h < H\}$. Its transform converges first in a left half-plane. If the allowed negative-end rate is b , the initial domain is $\text{Re } z < -b$ and the source extends to $\text{Re } z < -b + \gamma$. Given b_1 with $b - \gamma < b_1 < b$, choose a new contour with $-b_1 < \sigma < -b + \gamma$ and $\sigma \neq 0$. On the $(-l)$ th translate its integral is $O(e^{-\sigma l})$, whose squared norms become summable

after multiplication by $e^{-b_1 l}$ because $\sigma + b_1 > 0$. Thus moving the contour to the right improves the negative-end rate; the residue discussion is unchanged.

It remains to check the iteration, since γ is fixed. For a homogeneous perturbed equation, each improvement in s improves the source $-Bs$ by the same amount. Choose, for example, improvements smaller than $\gamma/3$ at each step, perturbing contour positions if necessary to avoid real part zero. Canonical sections have no residue, so iteration reaches every finite decay rate. Scalar sections can cross zero once, giving $s = c_{\pm} + r$ with r at a strictly negative allowed rate. If $c_{\pm} = 0$, the same homogeneous iteration continues indefinitely. If $c_{\pm} \neq 0$, then

$$(d + B\wedge)r = -Bc_{\pm};$$

only a finite decay rate for this remainder is claimed. Uniform local elliptic estimates give the stated decay of all its derivatives. $\square$

Remark 4.5. The continuation calculation also identifies the possible homogeneous modes. More generally, a pole of finite order at the unit multiplier would give a polynomial-times-equivariant homogeneous mode. Multiplication of the transform by a power of $e^z - 1$ removes that pole, so the corresponding section is killed by a power of $T^* - 1$. Such a nonconstant scalar mode would have a last nonconstant member v with $(T^* - 1)v$ a nonzero constant, excluded by Lemma 2.3. A canonical generalized mode would have a last nonzero invariant member, contradicting $H^0(X, K_X) = 0$. Thus the explicit simple-pole and removable-singularity calculations agree with the absence of these generalized modes.

Proposition 4.6 (Tail asymptotics). *Let E satisfy (4.3). A finite-growth holomorphic section of $K_Y \otimes E$ or $K_Y \otimes E^{-1}$ on a positive tail vanishes on a shorter positive tail. A finite-growth holomorphic section of E has, in the prescribed trivialization at either end on which it is defined, the expansion*

$$s = c_{\pm} + O(e^{-\delta|h|})$$

for some $\delta > 0$, with the same conclusion for every derivative of the error. On the positive end $c_+ = 0$ implies $s = 0$. Consequently the positive-end coefficient is injective on global weighted holomorphic sections of E .

Proof. Apply Lemma 4.4. The Chern curvatures of the metrics used here are bounded: those of the periodic bundles are bounded, and the connection coefficients and their first derivatives in the end frames are bounded by (4.3). Every canonical section, and a scalar-type section with zero positive coefficient, decays faster than every exponential on the positive tail. Lemma 4.3 makes it zero there. Unique continuation for a holomorphic section on the connected Y then gives the global injectivity assertion. The negative-end expansion uses no maximum principle at that end and makes no vanishing assertion there. $\square$

Positive-tail uniqueness now controls the kernel of every compact datum. To obtain such a datum holomorphically, we next solve the scalar top-degree equation with arbitrarily rapid decay.

4.3 Rapid scalar solutions and holomorphic compact data

Lemma 4.7. *For the scalar periodic Dolbeault complex and every admissible pair,*

$$H_{a,b}^{0,2}(Y, \mathcal{O}) = 0.$$

In particular, if f is a smooth compactly supported $(0, 2)$ -form and $R > 0$, there is a smooth $(0, 1)$ -form α with $\bar{\partial}\alpha = f$ and $\alpha \in W_{-R, -R}^k$ for every $k \in \mathbb{Z}$.

Proof. By Lemma 4.2 the dual of the group in question is $H_{-a,-b}^{2,0}(Y, \mathcal{O})$. Its elements are canonical sections with finite exponential growth on the positive end; they vanish by Proposition 4.6, applied to the periodic trivial datum. Closed range and finite-dimensionality turn the dual vanishing into the asserted vanishing. For f use the homotopy Q from (4.7) with weights $(-R, -R)$. The top-degree harmonic projection is zero, so $\alpha = Qf$ solves the equation. Elliptic regularity and the higher-order bounds for Q give all the asserted weighted Sobolev orders. $\square$

Proposition 4.8. *Every compact line-bundle datum admits an integrable $(0, 1)$ -connection satisfying (4.3) with any prescribed finite $\gamma > 0$. For every such structure and every admissible pair,*

$$H_{a,b}^{0,2}(Y, E) = 0. \quad (4.15)$$

For every $a, b > 0$, global weighted holomorphic sections have the two constant end coefficients and the exponentially decaying errors described in Proposition 4.6; the map $s \mapsto c_+$ is injective.

Proof. Choose a smooth $(0, 1)$ -connection d_0 on E which is standard in the given end trivializations. A partition of unity constructs one, because the space of such connections is affine. Its curvature d_0^2 is a compactly supported scalar $(0, 2)$ -form κ : the endomorphism bundle of a line bundle is canonically trivial. In complex dimension two it is automatically $\bar{\partial}$ -closed. Apply Lemma 4.7, already proved solely for the trivial periodic bundle, to solve

$$\bar{\partial}\alpha = -\kappa$$

with weights $(-R, -R)$, where $R > \gamma$. Define $d_E = d_0 + \alpha \wedge$. In a local frame its square is

$$d_E^2 = \kappa + \bar{\partial}\alpha + \alpha \wedge \alpha = 0.$$

The last term is zero because α is scalar-valued and has degree one. The local integrability theorem for a $(0, 1)$ -connection therefore makes d_E a holomorphic line-bundle structure. In rank one this theorem is just the local Dolbeault lemma applied to its closed connection form, followed by multiplication of the frame by the exponential of a local primitive. Weighted Sobolev embedding on the uniform charts gives $|\nabla^j \alpha| \leq C_j e^{-R|h|}$ on the tails, hence the prescribed bound with γ . The finitely many middle charts impose no end condition.

Now weighted duality identifies the dual of $H_{a,b}^{0,2}(Y, E)$ with $H_{-a,-b}^{2,0}(Y, E^{-1})$. The latter vanishes by Proposition 4.6. This proves (4.15). The section assertions follow from the same proposition, whose finite-growth hypothesis permits every positive finite rate. The scalar top-degree vanishing used to construct d_E preceded, and did not assume, the existence of this holomorphic structure. $\square$

The preceding solution makes the line-bundle data integrable with a prescribed end rate. Their top-degree cohomology is zero, and the positive-end coefficient bounds the dimension of their section spaces by one. The index calculation will show that this bound is attained for the classes in Lemma 3.1.

4.4 The index and compact excision

Lemma 4.9. *For the trivial datum and $a, b > 0$,*

$$H_{a,b}^{0,0}(Y, \mathcal{O}) = \mathbb{C}, \quad H_{a,b}^{0,1}(Y, \mathcal{O}) = H_{a,b}^{0,2}(Y, \mathcal{O}) = 0.$$

In particular its weighted Dolbeault index is one.

Proof. A global weighted holomorphic function has a positive-end constant coefficient. Subtract that constant; the resulting holomorphic function has zero positive coefficient and vanishes by Proposition 4.6. Constants belong to the space because both end rates are positive. The top-degree assertion is Lemma 4.7.

Here are the details in degree one. Represent a class by a smooth weighted harmonic $(0, 1)$ -form u , so all its weighted Sobolev norms are finite. Choose $\chi_+ + \chi_- = 1$ with χ_+ zero far down the negative end and one far up the positive end. The compactly supported $(0, 2)$ -form $g = \bar{\partial}(\chi_+ u) = -\bar{\partial}(\chi_- u)$ admits a primitive v decaying at both ends at any prescribed finite rate, by Lemma 4.7. Choose that rate larger than $a + b + 1$. Then

$$u_+ = \chi_+ u - v, \quad u_- = \chi_- u + v$$

are closed, with $u = u_+ + u_-$. The first belongs to the single-linear space $(a, -a)$, because its negative tail is $-v$; the second belongs to the single-linear space $(-b, b)$, because its positive tail is v . Both complexes are acyclic by Proposition 4.1. Choose $\bar{\partial}f_+ = u_+$ and $\bar{\partial}f_- = u_-$ in those respective spaces. Each of these spaces embeds into the two-growth-end space (a, b) : one end has the same weight and the other has a stronger decay condition. Thus $u = \bar{\partial}(f_+ + f_-)$ in the original complex. $\square$

Proposition 4.10. *Let E be a compact datum supported in a cut-open period of a finite cyclic quotient $X_n = Y/T^n$, with its trivializations fixed in collars of the cut. Equip E with an integrable operator d_E satisfying (4.3), and let e be its compact support Chern class. For $a, b > 0$ its weighted Dolbeault index is*

$$\sum_{q=0}^2 (-1)^q \dim H_{a,b}^{0,q}(Y, E) = 1 + \frac{1}{2}(e^2 - K_Y \cdot e). \quad (4.16)$$

The products on the right are compact support intersection pairings. In particular, if $e^2 = K_Y \cdot e$, the section space is one-dimensional, its positive-end coefficient is nonzero on every nonzero section, and $H_{a,b}^{0,1}(Y, E) = 0$.

Proof. The even-to-odd operator

$$D_E^+ = d_E + d_{E,\phi}^* : W_{a,b}^1(\Lambda^{0,0}E \oplus \Lambda^{0,2}E) \longrightarrow L_{a,b}^2(\Lambda^{0,1}E)$$

has index equal to the Euler characteristic in the statement, by the Hodge decomposition proved above. Replace d_E by any smooth $(0, 1)$ -connection d_0 standard at the ends. The difference, after conjugation, is multiplication by exponentially decaying coefficients, and is compact from W^1 to L^2 . The index is unchanged. We do not require $d_0^2 = 0$ for this replacement: D_0^+ is still a first-order elliptic operator with the same principal symbol and the same invertible end limits.

Transfer the datum and this connection to X_n , making them trivial and standard across the cut collars. Denote the resulting smooth bundle by E_n and its elliptic operator by $D_{E_n}^+$. Modify $\phi_{a,b}$ on a compact set so that it is zero on a neighborhood of the entire transferred piece and its collars; this does not change the index. The following excision uses the same compact cut-and-paste principle as [24, Proposition 3.1]; we give the operator comparison in the present setting:

$$\text{ind } D_{E,Y}^+ + \text{ind } D_{\mathcal{O},X_n}^+ = \text{ind } D_{\mathcal{O},Y}^+ + \text{ind } D_{E_n,X_n}^+. \quad (4.17)$$

To see it directly, compare these two direct sums. On the common compact interior interchange the two copies, carrying the E copy to the transferred E_n copy and the trivial copy to the trivial copy.

Outside the identified region use the identity. In each collar both bundles are trivialized; interpolate between interchange and identity using the real rotation matrix

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad 0 \leq \theta \leq \pi/2.$$

Use the same rotation on the even and odd bundles. This defines bounded invertible maps on the direct-sum domains and ranges. On the interior the operators are interchanged exactly. On the collars their principal symbols are the identical Dolbeault symbols, which commute with the rotation; differentiating the rotation only adds order-zero terms. Outside the compact comparison region they agree. The conjugated operators therefore differ by compactly supported order-zero terms, which are compact from W^1 to L^2 . Their indices agree, proving (4.17).

The transferred connection on X_n need not be integrable. To compute its index nevertheless by ordinary Hirzebruch–Riemann–Roch, recall from Lemma 2.5 that $H^2(X_n, \mathcal{O}_{X_n}) = 0$. The exponential sequence then shows that every integral class in $H^2(X_n; \mathbb{Z})$ is the first Chern class of a holomorphic line bundle. Smooth complex line bundles are classified by their first Chern class, so E_n admits some holomorphic structure. Replacing the transferred connection by its Dolbeault operator changes the closed elliptic operator only in order zero and hence preserves its index. Compact-complex Riemann–Roch [2, §4] gives

$$\operatorname{ind} D_{E_n, X_n}^+ - \operatorname{ind} D_{\mathcal{O}, X_n}^+ = \chi(X_n, E_n) - \chi(X_n, \mathcal{O}_{X_n}) = \frac{1}{2}(c_1(E_n)^2 - K_{X_n} \cdot c_1(E_n)).$$

These products equal e^2 and $K_Y \cdot e$, since the class was transferred in its fixed trivialized collars. Combining (4.17) with Lemma 4.9 proves (4.16).

When the correction term is zero, Proposition 4.8 gives $H^{0,2} = 0$ and Proposition 4.6 gives $\dim H^{0,0} \leq 1$. The index equality then reads $\dim H^{0,0} - \dim H^{0,1} = 1$. Both dimensions are nonnegative, so they are respectively 1 and 0, with the asserted nonzero positive coefficient. $\square$

4.5 Sections with prescribed end limits

For the compact classes constructed earlier the index correction vanishes, so a section with positive coefficient 1 now exists. We finish the construction by determining its negative coefficient and identifying the class of its divisor.

Proposition 4.11 (A section with different end limits). *Let E be one of the data in Lemma 3.1, and write $e^2 = K_Y \cdot e = -m$ with $m > 0$. Given any finite $\gamma > 0$, E admits a holomorphic structure whose Dolbeault operator, in its fixed end frames, differs from the trivial operator by $O(e^{-\gamma|h|})$ in every derivative. For this structure there is a holomorphic section s such that, in those frames,*

$$s \longrightarrow 1 \quad (h \rightarrow +\infty), \quad s \longrightarrow 0 \quad (h \rightarrow -\infty), \quad (4.18)$$

with exponentially decaying errors. Its effective divisor $D = (s = 0)$ is contained in $\{h \leq H\}$ for some H , is locally finite, and represents the ordinary Chern class $j(e)$, or equivalently its dual locally finite homology class.

Proof. Proposition 4.8 supplies the stated holomorphic structure with the prescribed finite rate γ . Choose any $\varepsilon > 0$ and use the pair $(\varepsilon, \varepsilon)$. Proposition 4.10 gives

$$\operatorname{ind} \bar{\partial}_E = 1 + \frac{1}{2}(e^2 - K_Y \cdot e) = 1.$$

By the weighted degree-two vanishing and the positive-end uniqueness in Propositions 4.8 and 4.6, $H_{\varepsilon, \varepsilon}^{0,2}(Y, E) = 0$ and the positive leading coefficient is an injective map

$$H_{\varepsilon, \varepsilon}^{0,0}(Y, E) \longrightarrow \mathbb{C}.$$

The index identity therefore implies that this degree-zero space is one-dimensional (and the degree-one space is zero). Take its nonzero section and normalize the positive leading coefficient to one. Proposition 4.6 gives a constant negative leading coefficient c_- and exponentially small errors at both ends.

Suppose $c_- \neq 0$. Uniform convergence to nonzero constants would make s nowhere zero outside a compact height slab. Its zero divisor D would then be compact. Moreover its compact divisor class would equal the particular relative class e , not just its ordinary image. To see the role of the framings, write $s = f_{\pm} \tau_{\pm}$ on the ends in the fixed frames. Far enough out, f_+ lies in a disk about 1 that avoids zero and f_- lies in a disk about c_- that avoids zero. Straight-line homotopies in these disks deform $f_{\pm}$ through nowhere-zero maps to their limiting constants. Each nonzero constant is in turn homotopic to 1 in $\mathbb{C}^*$. Thus the trivializations provided by s on the ends are homotopic to the prescribed ones. The relative zero-divisor formula for c_1 consequently identifies the compact divisor class with e .

The restriction of π to the compact divisor is proper. Its pushforward $\pi_* D$ is an effective divisor on X , with positive integral multiplicities equal to the corresponding generic mapping degrees. Since $K_Y = \pi^* K_X$, the projection formula gives

$$K_X \cdot \pi_* D = K_Y \cdot D = K_Y \cdot e = -m < 0.$$

This contradicts Lemma 2.2; if the divisor were empty, the left side would instead be zero and give the same contradiction. Therefore $c_- = 0$, proving (4.18).

The positive limit excludes zeros for all sufficiently large h . The zero set of a nonzero holomorphic section is a locally finite effective divisor. The local divisor construction identifies $E \simeq \mathcal{O}_Y(D)$, so $[D] = c_1(E) = j(e)$ in ordinary cohomology; the corresponding Chern-current identity is the local form of [7, Chapter V, (13.2)]. Under Poincaré duality for the oriented noncompact four-manifold Y , this says that its locally finite fundamental cycle is the dual of $j(e)$. In particular, if α is any compactly supported closed two-form, then

$$\int_Y \alpha \wedge j(e) = \int_D \alpha. \quad (4.19)$$

The right side is well-defined by local finiteness. No assertion that D is a compact divisor is being made here. $\square$

The rate γ in Proposition 4.11 is free. The next sections first obtain an area exponent A from the fixed lifted coordinate geometry, and then use this proposition with $\gamma > 2A + 10$. Thus the choice of that exponent imposes no additional condition on the compact classes constructed above.

5 Divisor growth and residue currents

Proposition 4.11 constructs a section with end limits $(1, 0)$ for any prescribed finite decay rate of the holomorphic structure. Before choosing that rate, we prove an area bound with an exponent A depending only on the lifted geometry. We then choose the holomorphic structures to decay faster than this common exponent.

Our goal is to constrain the normalizations of the resulting divisor components. A holomorphic differential on a normalization defines a closed current on Y . We place currents with exponential mass bounds in fixed weighted cohomology spaces and prove that currents carried by distinct curves have independent classes. The scalar degree-one group vanishes, so scalar differentials satisfying the mass bound cannot occur. This will force genus zero here; Section 6 will exclude the remaining noncompact normalizations, using the same scalar obstruction and the finite-dimensional cohomology for a fixed bundle E .

5.1 A common area exponent

Areas are measured with the lifted Hermitian metric; multiplicities are included when the argument is a divisor. Put

$$I_l = [-l - 1, -l], \quad Y_l = h^{-1}(I_l) \quad (l \in \mathbb{N} \cup \{0\}).$$

Lemma 5.1 (A geometric area exponent). *There is a number $A > 0$, depending only on a fixed finite coordinate atlas of X , its lifted metric, and the covering height, with the following property. Let E be any compact datum whose holomorphic operator tends smoothly to the trivial operator at the negative end, and let s be a nonzero holomorphic section which is bounded in the prescribed smooth trivialization there. If $D = (s = 0)$, then*

$$\text{Area}(D \cap Y_l) \leq C_{E,s} \exp(A l), \quad l \geq 0. \quad (5.1)$$

The constant $C_{E,s}$ may depend on the datum, operator, and section. The exponent A is independent of their decay rate. The same exponent applies to every reduced component of D and each of its fixed deck translates. Replacing I_l by any fixed-width enlargement also preserves the exponent.

Proof. Here are details of the quantitative propagation used in the proof. Choose a compact set $Q \subset Y$ whose deck translates cover Y . Choose finitely many small coordinate balls covering Q , and paths joining their centers to a point p , together with a path from p to $T^{-1}p$. The union of these paths is compact. Subdividing the paths and slightly enlarging the balls produces a finite collection of ball chains. Repeating its translates connects a starting ball to every ball covering $T^{-l}Q$ in at most $N(l+1)$ transitions, for one fixed N . All balls in such a chain lie in a fixed-width enlargement of the height interval traversed by the chain. These statements use only connectedness of Y , compactness of the chosen paths, and the identity $h \circ T = h + 1$; connectedness of individual height levels is unnecessary.

We recall the elementary analytic estimate for a transition. If F is holomorphic on a Euclidean ball of radius r_3 , and $0 < r_1 < r_2 < r_3$, the three-circle inequality is

$$\log \sup_{B_{r_2}} |F| \leq \vartheta \log \sup_{B_{r_1}} |F| + (1 - \vartheta) \log \sup_{B_{r_3}} |F|, \quad \vartheta = \frac{\log(r_3/r_2)}{\log(r_3/r_1)}. \quad (5.2)$$

Indeed, restricting to each complex line through the center gives the one-variable three-circle inequality, and then taking the supremum over the lines gives the displayed inequality. Choose the ball chains so that a small ball at one stage is contained in a middle ball at the next stage, while that stage's large ball remains inside its coordinate chart. This is possible by subdividing the paths further. Only finitely many radius ratios and coordinate changes are involved. Consequently the corresponding numbers ϑ have a positive lower bound ϑ_0 . The same construction, with a bounded number of additional transitions, reaches the small concentric balls needed for all the local integrations below.

On a sufficiently negative tail write the operator as $\bar{\partial} + \alpha$ in the prescribed smooth frame. Since $\alpha \rightarrow 0$ smoothly and $\bar{\partial}\alpha = 0$, on each of our large coordinate balls there is a function u with $\bar{\partial}u = \alpha$ and a uniform bound $|u| \leq C_0$ on a slightly smaller ball. This is the local Dolbeault lemma with its interior estimate. One can obtain the estimate directly by the Cauchy–Green operator in the first coordinate and then in the second: closedness removes the first antiholomorphic derivative from the remaining coefficient; the integrable kernels give a bound in terms of finitely many coefficient norms on the larger polydisc. The finite atlas makes these bounds uniform. Moving the start of the tail makes the coefficient norms at most one, independently of the original decay rate. The frames e^{-u} are holomorphic and their norms, and the absolute values of transition functions between these frames, are bounded above and below by geometric constants.

Let F denote the coefficient of s in such a holomorphic frame. The boundedness of s gives a common upper bound $\log |F| \leq U$ on the tail, where $U \geq 1$ may depend on s . At every transition, (5.2), the inclusion of the preceding small ball in the next middle ball, and the bounded frame changes show the following. If the logarithmic supremum on the preceding small ball is at least $-M$, then on the next small ball it is at least

$$-cM - C_1(U + 1), \quad c = \vartheta_0^{-1} > 1.$$

Increasing c if needed accommodates any transition with equal radii. Since s is nonzero, unique continuation supplies a finite starting value M . Start sufficiently far down the tail that all repeated chains stay in the region where the frame bounds hold. Iteration over at most $N(l + 1)$ transitions gives

$$\sup_V \log |F| \geq -C_2 c^{Nl} \tag{5.3}$$

on every small ball V needed to cover the l th negative cell. Here and below a change of the starting cell changes C_2 , not c or N . A slab Y_l meets only a fixed number of these cells, because h is bounded on Q .

For completeness the conversion from a logarithmic supremum to an L^1 bound is elementary. Arrange nested coordinate balls of radii $r, 2r, 4r$ and choose $x \in B_r$ with $\log |F(x)| \geq -M - 1$, using (5.3). The nonnegative function $v = U - \log |F|$ is superharmonic. The ball $B(x, 3r)$ contains B_{2r} and lies in B_{4r} , so its mean inequality gives

$$\int_{B_{2r}} v \leq \int_{B(x, 3r)} v \leq |B_{3r}| v(x) \leq |B_{3r}|(U + M + 1).$$

The usual limiting interpretation applies at zeros. Thus $\int_{B_{2r}} |\log |F|| \leq C_3(U + M + 1)$. Choose a smooth cutoff χ equal to one on B_r and supported in B_{2r} . Poincaré–Lelong [7, Chapter III, (2.15)] and distributional integration by parts give, up to the fixed normalization of d^c ,

$$\int_D \chi \omega = c_0 \int \log |F| dd^c(\chi \omega) \leq C_4 \int_{B_{2r}} |\log |F||.$$

The coefficients of $dd^c(\chi \omega)$ are uniformly bounded. No closedness of the Hermitian form ω is needed for this local identity. Summing over the fixed number of balls covering a slab and using (5.3) proves (5.1), for example with any fixed $A > N \log c$. Finitely many slabs outside the chosen tail are compact and are absorbed in $C_{E,s}$.

A reduced component has no more area than the effective divisor. Since T is an isometry and raises height by one, replacing a component by $T^k B$ shifts the slab index by k . After absorbing the finitely many remaining slabs in the constant, its exponent is still A . A fixed-width enlarged slab is contained in a bounded number of unit slabs whose indices differ from l by a bounded amount. This proves the final assertion as well. $\square$

From now on fix this geometric A , and make every application of Proposition 4.11 with a decay rate

$$\gamma > 2A + 10. \quad (5.4)$$

That proposition was proved for every prescribed finite rate, so this is a specialization of its construction. For each datum we obtain E, s, D with $s \rightarrow 1$ at the positive end and $s \rightarrow 0$ at the negative end. In particular D is bounded above in height. Lemma 5.1 applies to these choices with the already fixed exponent A .

5.2 Currents and their weighted cohomology

Let $B \subset Y$ be a closed irreducible analytic curve, possibly noncompact and bounded above in height, and let $f : R \rightarrow B \subset Y$ be its normalization. Normalization is finite over each relatively compact open subset [7, Chapter II, (7.12)], so f is proper. If F is $\mathcal{O}$ or one of the bundles E , and η is a holomorphic section of $K_R \otimes f^*F$, define its pushforward current $C_\eta = f_*\eta$ by

$$C_\eta(\varphi) = \int_R \eta \wedge f^*\varphi, \quad \varphi \in C_c^\infty(Y, \Lambda^{0,1} \otimes F^*). \quad (5.5)$$

The current has bidegree $(2, 1)$ and coefficients in F . Properness makes the integral locally finite, including at singular points of B . Stokes' theorem on R , applied to a compactly supported pullback, shows $\bar{\partial}_F C_\eta = 0$. With any conformal metric on R there is a fixed convention-dependent constant c such that

$$\text{Mass}(C_\eta; V) \leq c \int_{f^{-1}(V)} |\eta| |df| dA_R. \quad (5.6)$$

The expression on the right is conformally invariant. It also shows directly that C_η is of order zero.

We need three properties of these currents: finite-dimensional cohomology for a fixed coefficient bundle, vanishing of scalar cohomology in degree one, and vanishing on a positive tail of any $(2, 0)$ -primitive of a $(2, 1)$ -current supported below a height bound. The weighted results of Section 4 give all three in the following fixed spaces. The mass criterion at the end of the theorem specifies which currents belong to them.

Theorem 5.2 (Weighted cohomology for residue currents). *Fix $\varepsilon > 0$ and $b > \varepsilon$. Let F be either $\mathcal{O}_Y$ or an end-framed compact datum E with an integrable operator satisfying (4.3). Consider the current complex*

$$\mathcal{C}_{-\varepsilon, b, -3}^{2, q}(F) = W_{-\varepsilon, b}^{-3-q}(\Lambda^{2, q} \otimes F), \quad q = 0, 1, 2. \quad (5.7)$$

Its cohomology is finite-dimensional. For $F = \mathcal{O}_Y$ its degree-one cohomology is zero. In particular every closed scalar $(2, 1)$ -current in this space which is zero on a positive tail is exact.

If a degree-zero current $\beta \in W_{-\varepsilon, b}^{-3}(\Lambda^{2, 0} \otimes F)$ satisfies $\bar{\partial}_F \beta = J$ for a degree-one current J supported in $\{h \leq C\}$, then β vanishes on a positive tail. Finally, an order-zero $(2, 1)$ -current supported in $\{h \leq C\}$ whose masses on negative unit height slabs are at most $C_0 e^{\mu l}$ belongs to this complex whenever $b > \mu$. The same Sobolev order and pair of rates can be used for any family with that common exponent μ , regardless of its individual constants and upper support bounds.

Proof. Finite-dimensionality is Proposition 4.1 with $s = -3$. We first give the direct exactness argument needed for bounded-above support. If J is scalar and zero on the positive tail, it belongs to the single-linear-rate complex $(-b, b)$: its negative weight is unchanged and its positive part has compact support. That complex is acyclic. Its primitive has rates $(-b, b)$ and therefore also belongs to $(-\varepsilon, b)$, since $b > \varepsilon$.

The full scalar degree-one vanishing can also be checked without a contour argument in higher degree. Keep b fixed and continuously vary the positive rate from $-b$ to $-\varepsilon$. After conjugation the even-to-odd Hodge operators have the same domain W^1 and range L^2 , and vary continuously in operator norm: only bounded order-zero coefficients depending on the derivative of the weight change. They remain Fredholm by Proposition 4.1, since the positive rate stays negative. Their index is consequently constant, and is zero at the acyclic single-linear pair $(-b, b)$. At every point of this path, $H^{2,0} = 0$ by positive-tail uniqueness. By weighted duality $H^{2,2}$ is dual to the scalar holomorphic functions with a positive rate at the positive end and the negative rate $-b$ at the negative end. Every such global scalar function is constant: subtract its positive-end coefficient and use Proposition 4.6. The negative-end decay requirement excludes a nonzero constant. Thus $H^{2,2} = 0$ as well. Index zero now implies $H^{2,1} = 0$.

For the primitive assertion, on $\{h > C\}$ the distributional section β solves the degree-zero equation for $K_Y \otimes F$. Local elliptic regularity makes it smooth there. Uniform interior estimates on shorter tails turn its weighted W^{-3} bound into finite exponential growth in every positive Sobolev order. Proposition 4.6 therefore makes it zero on a positive tail. This holds for every primitive in the stated space.

To verify the last membership assertion, localize to the uniformly bounded coordinate charts on one enlarged slab. In real dimension four, W^4 embeds continuously into C^0 , with a uniform constant on these charts. An order-zero current of mass M thus has W^{-4} norm at most CM . A uniformly locally finite partition of unity and the bounded derivatives of the weight give an upper bound for the squared weighted norm by a constant multiple of

$$\sum_{l \geq 0} e^{-2bl} (C_0 e^{\mu l})^2 = C_0^2 \sum_{l \geq 0} e^{-2(b-\mu)l} < \infty.$$

There are only finitely many additional slabs between the zero level and the upper support bound, and none beyond it. This proves membership in $W_{-\varepsilon, b}^{-4}$, which is degree one of (5.7). The constants of individual currents enter their norms, not the choice of spaces or the dimension of their common cohomology group. $\square$

In particular, if an order-zero $(2, 1)$ -current C with coefficients in $F \in \{\mathcal{O}, E\}$ is bounded above in height and satisfies $\text{Mass}(C; Y_l) \leq C_C e^{\mu l}$, then for any $\varepsilon > 0$ and $b > \max\{\mu, \varepsilon\}$ the theorem places it in

$$W_{-\varepsilon, b}^{-4}(\Lambda^{2,1} \otimes F).$$

The inequality $b > \mu$ makes the weighted mass sum converge. The base Sobolev order remains -3 and the degree-one order remains -4 , independently of both the current and its mass exponent. We will always fix the rates before comparing current classes; no dimension bound uniform over changing weights is needed.

5.3 Residue independence

If a linear combination of our currents is exact, its primitive vanishes first on a positive tail and then off the supporting curves by analytic continuation. A local distribution supported on a curve cannot have the required order-zero residue as its Dolbeault derivative. This gives the independence statement that we will use both for scalar and for bundle-valued differentials.

Proposition 5.3 (Residue obstruction). *Fix $F \in \{\mathcal{O}, E\}$ and rates $(-\varepsilon, b)$ as in Theorem 5.2. For distinct closed irreducible curves $B_i \subset Y$ bounded above in height, let η_i be nonzero holomorphic*

differentials on their normalizations with coefficients in f_i^*F . Suppose their pushforward currents have negative-end mass exponents strictly smaller than b . Then the classes

$$[C_{\eta_i}] \in H_{-\varepsilon, b}^{2,1}(Y, F)$$

are linearly independent: every finite subfamily is independent. For $F = \mathcal{O}$, even one such differential is impossible.

Proof. The mass criterion in Theorem 5.2 puts the currents in the indicated complex. Suppose a finite linear combination is exact:

$$\sum_{i=1}^r a_i C_{\eta_i} = \bar{\partial}_F \beta.$$

The sum is zero on a positive tail. The primitive assertion in Theorem 5.2 says that β is zero there as well. On $Y \setminus \bigcup_i B_i$, its distributional Dolbeault derivative vanishes. Local elliptic regularity makes it a holomorphic section of $K_Y \otimes F$ on that complement. The complement is connected: a path in the connected real four-manifold can be perturbed off the smooth real-codimension-two strata of a finite union of analytic curves, and off its locally discrete singular strata. Only finitely many charts are needed along the compact path. Analytic continuation from the positive tail therefore gives

$$\text{supp } \beta \subset \bigcup_{i=1}^r B_i.$$

Suppose $a_i \neq 0$. Choose a smooth point of B_i , outside the other curves, where $\eta_i \neq 0$. Such a point exists since the singular and intersection sets are locally discrete and a nonzero holomorphic differential has isolated zeros. In holomorphic coordinates (z, w) and a holomorphic frame of F , the curve is $w = 0$, the differential is $a(z) dz$ with $a \not\equiv 0$, and $\beta = u dz \wedge dw$ for a distribution supported on $w = 0$. Its local finite normal-jet expansion has the form

$$u = \sum_{p+q \leq N} u_{pq}(z) \partial_w^p \bar{\partial}_w^q \delta_0(w), \quad (5.8)$$

with distributional tangential coefficients. This is the support theorem for distributions [19, Theorem 2.3.5]; locally it also follows by taking a finite-order bound for u , Taylor-expanding a test function in its normal variables, and cutting off in $|w| < \delta$. A test function whose normal derivatives through order N vanish on $w = 0$ has C^N norm $O(\delta)$ after that cutoff, and hence pairs to zero as $\delta \rightarrow 0$. Thus only the displayed finite Taylor jet can matter. The normal jets in (5.8) are independent, as is seen by prescribing those jets of test functions.

The coefficient of the normal exterior-form direction in $\bar{\partial}\beta$ is $\partial_{\bar{w}}u$; its tangential direction cannot contribute to that coefficient. The corresponding coefficient of the right-hand side is a nonzero constant multiple of $a_i a(z) \delta_0(w)$. But normal differentiation in (5.8) increases q by one. Comparing the largest $q + 1$, then descending, forces every u_{pq} to vanish and leaves zero in place of this nonzero order-zero delta mass. This contradiction proves independence. Notice that the familiar principal-value meromorphic primitive of a residue is not supported on the divisor; the support conclusion above is essential.

For $F = \mathcal{O}$, Theorem 5.2 makes every closed current of the kind under consideration exact: its support is bounded above, so a single linear weight $(-b, b)$ supplies the scalar primitive. The independence assertion then rules out even one nonzero current. $\square$

5.4 The topology of the normalization

An L^2 differential satisfies the required current mass bound by Cauchy–Schwarz and the area estimate. Thus the scalar residue obstruction excludes such differentials. The next proposition turns this analytic conclusion into a restriction on the topology of the normalization, including when it is noncompact.

Proposition 5.4 (Genus zero). *The normalization R of any component of D has no nonzero square-integrable scalar holomorphic differential and has genus zero. If R is compact, it is biholomorphic to $\mathbb{P}^1$. If it is noncompact, it is biholomorphic to a domain in $\mathbb{P}^1$, without a restriction on the connectivity of that domain.*

Proof. Let η be a square-integrable holomorphic differential on R . Cauchy–Schwarz in (5.6) gives

$$\text{Mass}(C_\eta; Y_l) \leq c \|\eta\|_{L^2(R)} \left(\int_{f^{-1}(Y_l)} |df|^2 dA_R \right)^{1/2} \leq C \|\eta\|_{L^2(R)} e^{Al/2}.$$

Here the energy of the holomorphic normalization is a fixed multiple of the reduced curve area; ramification and singular points are measure-zero sets. Lemma 5.1 gives the last bound. The scalar case of Proposition 5.3 forces $\eta = 0$.

We give the topological consequence also for open R . A handle in an oriented surface contains two embedded circles with algebraic intersection one. Smooth closed forms α, β , supported in small annular neighborhoods of these circles and dual to them, have

$$\int_R \alpha \wedge \beta = 1.$$

For example a bump function of the transverse coordinate, times its differential and with integral one, gives each such dual form. Choose any smooth conformal metric and let V be the closure of $dC_c^\infty(R, \mathbb{R})$ in the real Hilbert space of L^2 one-forms. Set $u = \alpha - \text{proj}_V \alpha$. Forms in V are weakly closed, so $du = 0$; orthogonality to dC_c^∞ gives $d^*u = 0$. Integration by parts against the compactly supported closed form β , followed by L^2 continuity, gives

$$\int_R u \wedge \beta = \int_R \alpha \wedge \beta = 1.$$

Thus $u \neq 0$. Local elliptic regularity makes u smooth. With the complex orientation, $u + i * u$ is a nonzero $(1, 0)$ -form; the two equations for u make it closed and hence holomorphic. It is L^2 , a contradiction. No closed-range assertion for the de Rham differential on a noncompact surface has been used.

There is consequently no handle in any compact subsurface of R . This is genus zero. Equivalently, every Jordan curve separates: a nonseparating Jordan curve can be joined from one side to the other in its complement, producing a second circle with intersection one and hence a handle in a compact neighborhood. A compact genus-zero Riemann surface is $\mathbb{P}^1$. For noncompact R , apply Koebe’s general planar uniformization theorem in the form [18, Theorem 1.1]: every planar Riemann surface admits a conformal embedding into the Riemann sphere. The criterion that every Jordan curve separates is its planarity hypothesis. The theorem allows arbitrary connectivity, so no finite-type assumption on R is being added. Its image is open, and thus is the asserted domain. $\square$

6 A cylinder estimate and compactness of the components

The normalizations of the divisor components have genus zero by Proposition 5.4. We prove a cylinder estimate and use it twice to exclude noncompact normalizations. First, on a planar domain omitting at least two points of the sphere, the estimate controls a nonzero scalar differential, contradicting the scalar residue obstruction. The only remaining open normalization is the plane. There a periodic gauge lets us apply the same estimate to bundle-valued differentials on infinitely many translated curves, contradicting finite-dimensionality of their fixed cohomology space.

The estimate controls the integral of a differential over a specified target height slab by the curve's area in a fixed enlargement of that slab. Its constants must remain bounded along arbitrarily long chains of low-energy source cylinders.

We retain the exponent $A > 0$ of Lemma 5.1. It was fixed from the coordinate geometry before the holomorphic structure on E was chosen. Its correction exponent satisfies $\gamma > 2A + 10$, as prescribed in (5.4). The normalization of any component of D , and of any fixed deck translate of that component, satisfies

$$\text{Area}(B \cap h^{-1}([-\ell - 1, -\ell])) \leq C_B e^{A\ell} \quad (\ell \in \mathbb{N}). \quad (6.1)$$

The constant C_B may depend on the translate. The exponent does not.

6.1 Estimates on source cylinders

Write $\mathcal{C} = \mathbb{R} \times (\mathbb{R}/2\pi\mathbb{Z})$, with coordinate $v = t + i\theta$ and flat area $dA_v = dt d\theta$. Put

$$S_j = [j, j+1] \times (\mathbb{R}/2\pi\mathbb{Z}), \quad C_j^r = (j-r, j+1+r) \times (\mathbb{R}/2\pi\mathbb{Z}).$$

Strip boundaries have zero area and can be included when specifying a point that a strip contains. For a holomorphic map f into Y let $u(v) = |df_v(\partial_t)|$. Conformality gives

$$f^*\omega = u^2 dA_v, \quad (6.2)$$

where ω is the Hermitian area form on complex tangent lines. Thus u differs from the Hilbert–Schmidt norm $|df|$ only by the fixed factor $\sqrt{2}$. We use u in the estimates to avoid this factor.

A finite nested coordinate cover of X , lifted to Y and recentered inside its smaller members, gives a radius $r > 0$ and centered holomorphic coordinate balls $U_p(4r)$ about every $p \in Y$. Fix these charts so that the Euclidean and lifted tangent norms are comparable with one constant $K \geq 1$. Decreasing r once if necessary makes all the recentered balls lie in the larger original charts. The smaller concentric balls $U_p(r)$ will be used in the definition of a flat strip.

Lemma 6.1 (Small energy fills a source cylinder). *There are an integer $P \geq 2$ and positive constants $Q, \delta, \varepsilon_0, D_0$, depending only on the lifted geometry, with the following property. Suppose that $\Omega \subset \mathcal{C}$ is a domain and $f : \Omega \rightarrow Y$ is a nonconstant proper holomorphic map. A strip S_j is called flat if $\overline{C_j^{2P+2}} \subset \Omega$ and the image of this entire thickened cylinder under f is contained in $U_p(r)$ for one of the fixed coordinate charts. Every flat strip satisfies $\sup_{S_j} u \leq D_0$.*

For every interval $I \subset \mathbb{R}$, if S_j contains a point v_j with $h(f(v_j)) \in I$ and S_j is not flat, then

$$\int_{\Omega \cap C_j^Q \cap (h \circ f)^{-1}(I + [-\delta, \delta])} u^2 dA_v \geq \varepsilon_0. \quad (6.3)$$

The same assertion holds on a source half-cylinder whenever $\overline{C_j^Q}$ lies in its interior.

Proof. Choose δ strictly larger than twice the maximum height oscillation of the balls $U_p(4r)$, with a fixed positive margin; a uniform multiple of $8Kr\|dh\|_\infty + 1$ suffices after increasing the chart comparison constant.

Choose $P \geq 2$ large enough that

$$q := \frac{2K^2}{e^P - 1} \leq \frac{1}{2}, \quad (6.4)$$

and set $Q = 3P + 4$. Cauchy's estimate on source disks of radius one, inside C_j^{2P+2} , gives a uniform bound D_0 for u on $\overline{C_j^{2P}}$ whenever S_j is flat. The norm of a coordinate map with image in $U_p(r)$ is at most r , so this bound depends only on K, r , and the fixed disk radius. In particular it bounds u on S_j .

It remains to prove a uniform positive energy cost. The issue is that the source domain may omit part of the thickened cylinder. We show that a sequence with energy tending to zero has a horizontal graph limit; proper projection near that limit then forces the missing source points to be present. If no ε_0 works, there are maps f_ν , nonflat strips, and marked points as in the statement for which the integral in (6.3) tends to zero. Translate source indices to $j = 0$ and translate the targets and their height intervals by the same powers of T so that the marked images lie in a fixed compact set. After a subsequence, the marked source points tend to $v_\infty \in S_0$ and their images tend to $p \in Y$. Denote the corresponding translated intervals by I_ν ; their lengths play no role in what follows. Work in the fixed coordinate ball $U_p(4r)$. For large ν , the marked image is arbitrarily close to p , so the strict margin in δ makes every point of this ball have height in the corresponding $I_\nu + [-\delta, \delta]$, even when the marked height lies at an endpoint of I_ν . The energy of the graph portions in $C_0^Q \times U_p(4r)$ therefore tends to zero in the target direction.

These portions

$$\Gamma_\nu = \{(v, f_\nu(v)) : v \in \Omega_\nu \cap C_0^Q, f_\nu(v) \in U_p(4r)\}$$

are closed analytic curves in that product. To check closedness, suppose $(v_k, f_\nu(v_k))$ converges in a compact subset of the product. Properness of f_ν puts the v_k in a compact subset of Ω_ν ; their limit is consequently in its domain and belongs to the graph. On a half-cylinder obtained by cutting a proper full normalization, the same proof works on compact source disks away from its artificial finite boundary; properness is applied to the ambient map. This is the place where properness is essential.

The product Hermitian area of a graph is the sum of its horizontal and vertical areas. Its horizontal area is at most $\text{Area}(C_0^Q)$, because the source projection has at most one point in each fiber. Its vertical area tends to zero. Bishop's compactness theorem [4], in the analytic-cycle and support formulation of [17, Theorems 3.9–3.10, pp. 585–586], therefore gives a subsequence converging locally as positive analytic cycles, and with convergence of their supports, to a nonempty pure one-dimensional analytic cycle Γ_∞ through (v_∞, p) . These theorems apply in local product coordinates (equivalently, identify the finite cylinder with an annulus). The locally bounded graph areas give their mass hypothesis. Convergence of positive currents makes the limit's vertical area zero. On the regular part of each component this says that the target coordinate functions have zero derivative, so each component is horizontal. Analytic continuation in the connected cylinder then gives

$$C_0^Q \times \{p\} \subset \text{supp } \Gamma_\infty.$$

For completeness, horizontal components cannot disappear at an interior source point: the defining holomorphic functions of the limit, restricted to the horizontal slice, vanish on an open set and hence on the whole connected slice.

We spell out why convergence to this slice fills a smaller cylinder. Fix a source disk $V \Subset C_0^Q$. Choose a small target ball $B(p, \eta) \Subset U_p(r)$ whose closure meets no other horizontal component of

the limit. Such a ball exists by local finiteness of the analytic cycle. On a slightly larger source disk the approximating graphs eventually avoid the target boundary $\partial B(p, \eta)$: otherwise a convergent sequence of graph points would give a limit point on that boundary. Thus the projection of the graph portion inside $V \times B(p, \eta)$ is proper. It is finite, since its fibers have at most one point, and it has nonempty image by convergence to the horizontal slice. A nonconstant finite holomorphic map of analytic curves is open, and a proper map is closed; its image is therefore all of the connected disk V . Its degree is exactly one, by the same single-point fiber bound. In particular every point of V belongs to Ω_ν , and its image is in $B(p, \eta)$. If there were a second horizontal component, the same argument in a disjoint target ball would give two graph points over one source point. Thus there is no second component.

Cover $\overline{C_0^{2P+2}}$ by finitely many such disks. It follows that for all sufficiently large ν this entire compact cylinder belongs to Ω_ν and its image is in $U_p(r)$. Interior Cauchy estimates also give convergence of its coordinate derivatives to zero on smaller cylinders. In particular S_0 is flat, contradicting its choice. This proves (6.3) and all the asserted uniformities. $\square$

Nonflat strips therefore cost a fixed amount of area near any marked height. For the flat strips we use a different bound: along an arbitrarily long consecutive chain, the sum of the derivative suprema remains bounded. This will also control the variation of target height.

Lemma 6.2 (Flat chains). *Let $J \subset \mathbb{Z}$ be a maximal interval of flat indices for a map as in Lemma 6.1. Put $m_j = \sup_{S_j} u$. There is a constant L_0 , depending only on the fixed geometry, such that*

$$\sum_{j \in J} m_j \leq L_0. \quad (6.5)$$

A doubly infinite chain is impossible for a nonconstant map. If a chain contains a point with height in an interval I , every finite endpoint of the chain has a neighboring nonflat strip containing a point with height in $I + [-\Lambda, \Lambda]$, where $\Lambda = \|dh\|_\infty L_0$ is fixed.

Proof. Take a flat strip S_j , and express f in its coordinate chart as a vector-valued periodic holomorphic function $F(v)$. Write $H = \partial_v F$. Since F is single valued around the cylinder,

$$\int_0^{2\pi} H(t + i\theta) d\theta = \frac{1}{i} \int_0^{2\pi} \partial_\theta F(t + i\theta) d\theta = 0.$$

Consequently the Laurent–Fourier expansion is $H(v) = \sum_{n \neq 0} c_n e^{nv}$. Bound the positive coefficients on the circle $t = j + 1 + P$ and the negative coefficients on $t = j - P$. For $j \leq t \leq j + 1$ their geometric sums give

$$|H(t + i\theta)| \leq \frac{2}{e^P - 1} \max \left\{ \sup_{t=j-P} |H|, \sup_{t=j+1+P} |H| \right\}.$$

The statement is valid for vector values by taking norms in the Fourier coefficient integrals. Comparing the coordinate and target norms yields

$$m_j \leq q \max_{|k-j| \leq P} m_k \quad (6.6)$$

whenever all the indices on the right belong to the chain. Each application uses the chart for its own central strip; the numbers m_k are intrinsic, so changing charts causes no further loss.

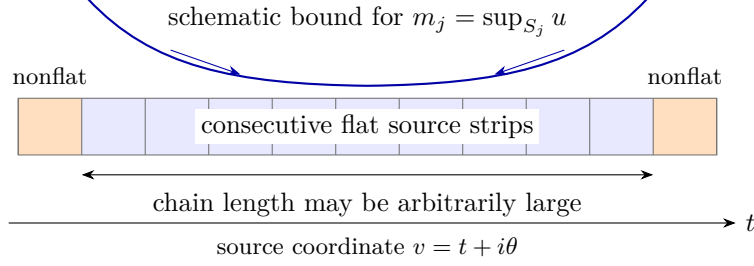


Figure 1: A flat chain in the source cylinder. The absence of a constant angular mode makes derivative bounds decay away from the chain's ends. Their sum, and hence the variation of target height along the chain, is bounded independently of its length. A chain meeting a target height slab can therefore be charged to a neighboring nonflat strip in a fixed enlarged slab. The horizontal coordinate is the source coordinate t , not the target height $h \circ f$.

All m_j in the chain are at most D_0 . Iterating (6.6) inside J therefore bounds m_j by $D_0 q^{\lfloor d(j)/P \rfloor}$, where $d(j)$ is the integer distance to the nearer finite end of J . On a half-infinite interval use its single finite end. At most $2P$ indices have any prescribed value of $\lfloor d(j)/P \rfloor$, so one may take

$$L_0 = \frac{2PD_0}{1-q}.$$

If $J = \mathbb{Z}$, the same iteration is possible arbitrarily often at every index and gives $df = 0$; analytic continuation contradicts nonconstancy.

Starting at a marked point of the chain, move in the t direction at its fixed angular coordinate toward either finite endpoint. The change of $h \circ f$ is at most $\|dh\|_\infty \sum_{j \in J} m_j \leq \Lambda$. The shared boundary with the neighboring nonflat strip is in the domain because the endpoint flat strip has a full thickening. That boundary point belongs to the neighboring closed strip and has height in $I + [-\Lambda, \Lambda]$. This proves the charging assertion. Figure 1 illustrates the derivative decay along the chain and the charge to a neighboring nonflat strip. $\square$

Proposition 6.3 (The cylinder estimate on a target height slab). *There are constants $R_0, C_0 > 0$ depending only on the lifted geometry with the following property. Let $f : \Omega \rightarrow Y$ be a nonconstant proper holomorphic map from a domain in $\mathcal{C}$. Let $w \geq 0$ be measurable and suppose*

$$M_w := \sup_{j \in \mathbb{Z}} \|w\|_{L^2(\Omega \cap S_j)} < \infty.$$

For every interval I of length one, set

$$E_I = \int_{\Omega \cap (h \circ f)^{-1}(I + [-R_0, R_0])} u^2 dA_v.$$

If $E_I < \infty$, then

$$\int_{\Omega \cap (h \circ f)^{-1}(I)} u w dA_v \leq C_0 M_w (1 + E_I). \quad (6.7)$$

The same estimate holds on a full source half-cylinder cut from a proper normalization map; properness is used only on thickened strips away from the artificial initial boundary, and a fixed finite initial block of strips is included in the constant. The integral on the left is always restricted to the inverse image of the target interval I .

Proof. Take $R_0 = \Lambda + \delta$, with constants from the two preceding lemmas. Let $\mathcal{N}_I$ be the set of nonflat strips containing a point of height in $I + [-\Lambda, \Lambda]$. The source cylinders C_j^Q have overlap at most $2Q + 3$. Summing (6.3) over any finite subset of $\mathcal{N}_I$ gives

$$\#\mathcal{N}_I \leq (2Q + 3)\varepsilon_0^{-1}E_I. \quad (6.8)$$

This also proves that the set being counted is finite. In particular, the set $\mathcal{N}'_I$ of nonflat strips hitting I has this bound. Cauchy–Schwarz, first on each strip restricted to the target slab and then on the finite sum, gives

$$\begin{aligned} & \sum_{j \in \mathcal{N}'_I} \int_{\Omega \cap S_j \cap (h \circ f)^{-1}(I)} u w dA_v \\ & \leq M_w (\#\mathcal{N}'_I)^{1/2} \left(\sum_{j \in \mathcal{N}'_I} \int_{\Omega \cap S_j \cap (h \circ f)^{-1}(I)} u^2 dA_v \right)^{1/2} \leq C M_w E_I. \end{aligned}$$

Only the intervals appearing explicitly in these integrals are used; no estimate of the energy outside the enlarged target interval is being assumed.

On one flat chain, (6.5) and $\|w\|_{L^1(S_j)} \leq \sqrt{2\pi}M_w$ give

$$\sum_{j \in J} \int_{S_j} u w dA_v \leq \sqrt{2\pi}M_w L_0.$$

Every chain meeting I has a finite endpoint adjacent to a member of $\mathcal{N}_I$, by Lemma 6.2. Each nonflat strip has at most two adjacent chains. Thus at most $2\#\mathcal{N}_I$ chains contribute, and their integral over the target slab is bounded by their full chain integrals. Combining this bound with (6.8) proves (6.7) on the full cylinder.

For a half-cylinder, omit the fixed finite block of at most $2Q + 4$ strips on which the small-energy neighborhoods meet its initial boundary. On this block Cauchy–Schwarz gives $CM_w E_I^{1/2} \leq CM_w(1 + E_I)$. Apply the preceding argument to the remaining indices. There is at most one chain reaching the new finite endpoint, and its integral is at most $\sqrt{2\pi}M_w L_0$. All other chains are charged as before. Increasing C_0 proves the half-cylinder assertion. $\square$

For a proper normalization the integral in (6.2) counts the area of the reduced curve once, away from its discrete singular set. A fixed enlargement of a height interval is covered by a fixed finite number of unit height intervals. Consequently (6.1) and Proposition 6.3 imply, with the same exponent A ,

$$\int_{(h \circ f)^{-1}([- \ell - 1, -\ell])} u w dA_v \leq C_{B,w} e^{A\ell} \quad (\ell \in \mathbb{N}). \quad (6.9)$$

On a half-cylinder the integral is also restricted to that end. Finitely many height intervals outside the range where the area estimate was initially proved change only the constant.

6.2 Excluding nonplane normalizations

We first apply the estimate with the constant multiplier $w = 1$. This supplies a differential with controlled current mass whenever the planar normalization omits at least two points of the sphere.

Corollary 6.4. *If a component of D is noncompact, its normalization is biholomorphic to $\mathbb{C}$.*

Proof. By Proposition 5.4, the normalization R is a domain in $\mathbb{P}^1$. If it omits two distinct points, send them by a Möbius transformation to 0 and ∞ . Then $R \subset \mathbb{C}^*$ is a domain in the logarithmic cylinder and $\alpha = dz/z = dv$ is a globally defined nonzero holomorphic differential on R . Formula (6.9) with $w = 1$ bounds the mass of $f_*\alpha$ on every unit height interval by $C_B e^{A\ell}$: pulling back a test one-form of norm at most one costs at most a fixed multiple of u , while $|dv|$ is constant. The scalar case of Proposition 5.3 forbids this nonzero differential. Thus $\mathbb{P}^1 \setminus R$ has at most one point. It has at least one point because R is noncompact. Removing that single point gives $R \simeq \mathbb{C}$. $\square$

6.3 Excluding plane normalizations

The scalar residue test leaves only plane normalizations. On a translated plane the differential dz has an exponentially growing coefficient in the logarithmic coordinate. We multiply it by the section of E that tends to zero at the negative end. To control the product, we first solve the pulled-back Dolbeault equation in a periodic gauge whose exponential factors have bounded L^2 norms on source strips. The next lemma supplies this gauge, including its angular constant mode. Its input a is an ordinary complex-valued function; in the application it will be the coefficient of the pulled-back connection.

Lemma 6.5 (A periodic gauge on the cylinder). *If $a \in L^1(\mathcal{C}) \cap L^2(\mathcal{C})$, there is a periodic distribution g satisfying*

$$\frac{1}{2}(\partial_t + i\partial_\theta)g = a, \quad \sup_{j \in \mathbb{Z}} \|g\|_{W^{1,2}(S_j)} \leq C(\|a\|_{L^1} + \|a\|_{L^2}).$$

For every $p < \infty$ this solution satisfies

$$\sup_j \int_{S_j} \exp(p|\operatorname{Re} g|) dA_v < \infty. \quad (6.10)$$

If a is smooth in an open set, g is smooth there.

Proof. Let $a_0(t) = (2\pi)^{-1} \int_0^{2\pi} a(t, \theta) d\theta$. Its primitive

$$g_0(t) = 2 \int_{-\infty}^t a_0(s) ds$$

is bounded by $\pi^{-1}\|a\|_{L^1}$, and $g'_0 = 2a_0 \in L^2(\mathbb{R})$. This bounds its full $W^{1,2}$ norm on each unit strip, including its mean. For $a_\perp = a - a_0$, take Fourier series in θ and Fourier transform in t . For every integer $n \neq 0$ define

$$\widehat{g}_\perp(\xi, n) = \frac{2\widehat{a}_\perp(\xi, n)}{i\xi - n}.$$

Since $|n| \geq 1$, each of the multipliers $1/(i\xi - n)$, $\xi/(i\xi - n)$, and $n/(i\xi - n)$ is bounded in absolute value by one. Plancherel therefore gives $\|g_\perp\|_{W^{1,2}(\mathcal{C})} \leq C\|a\|_{L^2}$. The sum $g = g_0 + g_\perp$ has the asserted periodicity, equation, and stripwise bound. Local elliptic regularity for $\partial_t + i\partial_\theta$ gives the last assertion.

Let B be a uniform bound for these stripwise norms. The two-dimensional Trudinger inequality on a fixed bounded strip [31], including the L^2 term in the norm, gives constants $c, C > 0$ independent of j such that

$$\int_{S_j} \exp\left(\frac{c|\operatorname{Re} g|^2}{B^2}\right) dA_v \leq C$$

when $B > 0$. One can cut the cylinder at a fixed angle and apply the bounded-rectangle form of that inequality; no boundary value of g is prescribed. The zero-norm case is immediate. For each

finite p , the inequality $p|x| \leq cx^2/B^2 + p^2B^2/(4c)$ proves (6.10). In particular both $e^{\operatorname{Re} g}$ and $e^{-\operatorname{Re} g}$ have bounded L^p norms on unit strips, for every fixed finite p . $\square$

Theorem 6.6. *Every irreducible component of D is compact and has rational normalization.*

Proof. It remains, by Proposition 5.4 and Corollary 6.4, to exclude a component with proper normalization $f : \mathbb{C} \rightarrow B \subset Y$. Let $M_B = \sup_B h < \infty$ and $M_D = \sup_D h < \infty$. The curves $B_k = T^k B$ are distinct, since their height suprema are $M_B + k$. For every sufficiently large positive integer k this supremum exceeds M_D , so B_k is not contained in D . Write $f_k = T^k \circ f$. We will construct on each of these curves the nonzero $f_k^* E$ -valued holomorphic differential

$$\alpha_k = (f_k^* s) dz \quad (6.11)$$

and prove that its pushforward has growth exponent A .

The key point is the cancellation of the growth of dz . Properness sends the exterior of the plane to the negative end of Y . After a periodic change of frame there, the section coefficient becomes an ordinary holomorphic function tending to zero at infinity, hence of size $O(1/|z|)$. Since $dz = z dv$, this leaves only an exponential gauge factor, whose stripwise L^2 bound is exactly the multiplier hypothesis in Proposition 6.3.

Fix one such k temporarily. Properness, together with the upper height bound, implies

$$h(f_k(z)) \longrightarrow -\infty \quad \text{uniformly as } |z| \longrightarrow \infty. \quad (6.12)$$

Indeed, otherwise a sequence with $|z| \rightarrow \infty$ would have image in some compact height slab $\{c \leq h \leq M_B + k\}$, contradicting properness. On an exterior disk $|z| > R_k$ we may therefore use the prescribed smooth negative-end frame of E . Choose its Hermitian norm to be one on that end. In this frame write $\bar{\partial}_E = \bar{\partial} + \eta$, where $|\eta| \leq Ce^{-\gamma|h|}$; this is the choice (5.4). In the logarithmic coordinate $v = \log z$, write $f_k^* \eta = a(v) d\bar{v}$. Then

$$|a(v)| \leq Ce^{-\gamma|h(f_k(v))|} u(v). \quad (6.13)$$

After increasing R_k , the entire image of this exterior disk has negative height. Summing (6.13) over its target height intervals, using (6.9) with $w = 1$ for the first estimate and (6.2) for the second, gives

$$\begin{aligned} \int |a| dA_v &\leq C_k \sum_{\ell \geq 0} e^{-(\gamma-A)\ell} < \infty, \\ \int |a|^2 dA_v &\leq C_k \sum_{\ell \geq 0} e^{-(2\gamma-A)\ell} < \infty. \end{aligned}$$

An initial compact annulus only adds a finite quantity. Smoothly cut a off toward the finite side and extend by zero to the full cylinder. This retains both bounds and agrees with a on a slightly smaller exterior disk. Lemma 6.5 gives a single-valued periodic smooth gauge g there, with $\bar{\partial}g = a d\bar{v}$ and uniformly bounded stripwise $W^{1,2}$ norms.

Let $\sigma(v)$ be the coefficient of $f_k^* s$ in the same smooth frame. Holomorphicity says $\bar{\partial}\sigma + a\sigma d\bar{v} = 0$. Hence

$$H(v) = e^{g(v)} \sigma(v)$$

is ordinary holomorphic and periodic on the exterior cylinder. By Proposition 4.11 and (6.12), $\sup_{S_j} |\sigma| \rightarrow 0$ as $j \rightarrow \infty$. The exponential estimate (6.10) therefore yields

$$\int_{S_j} |H|^2 dA_v \leq \left(\sup_{S_j} |\sigma|^2 \right) \int_{S_j} e^{2\operatorname{Re} g} dA_v \longrightarrow 0.$$

The mean-value inequality on fixed small disks in the cylinder, each contained in the union of three adjacent strips, implies $H(v) \rightarrow 0$ uniformly as $t \rightarrow \infty$. Viewed as a holomorphic function of z , it consequently extends holomorphically across $z = \infty$ with value zero. Its Taylor expansion in $1/z$ gives

$$|H(z)| \leq C_k |z|^{-1} \quad (6.14)$$

on a further exterior disk.

Since $dz = e^v dv$, Equations (6.11) and (6.14) give the coefficient estimate

$$|\alpha_k|_v = |e^v \sigma(v)| \leq C_k e^{-\operatorname{Re} g(v)}.$$

Here $|\cdot|_v$ is the norm of its coefficient relative to dv and the unit smooth frame. The right-hand factor has uniformly bounded L^2 norms on unit strips by (6.10). Apply (6.9) with $w = e^{-\operatorname{Re} g}$. Pullback of a test one-form costs at most Cu , so this proves

$$\operatorname{Mass} \left((f_k)_* \alpha_k \mid_{h^{-1}([- \ell - 1, -\ell])} \right) \leq C'_k e^{A\ell} \quad (\ell \geq 0). \quad (6.15)$$

The omitted compact disk in the source has compact image and contributes only on finitely many height intervals, which are absorbed in C'_k . The section $f_k^* s$ is nonzero because $B_k \not\subset D$, and dz is nowhere zero; thus α_k is a nonzero holomorphic bundle-valued differential.

Finally fix, once and for all, an admissible $\varepsilon > 0$ and a number $b > \max\{A, \varepsilon\}$. The currents in (6.15) all belong to the same degree-one space $W_{-\varepsilon, b}^{-4}$ of the shifted complex from Theorem 5.2; their coefficients take values in the same bundle E . Their positive supports are bounded separately, and any finite collection has a common upper support bound. Proposition 5.3 therefore makes the classes of every finite collection linearly independent in the single space $H_{-\varepsilon, b}^{2,1}(Y, E)$. That space is finite dimensional by Theorem 5.2. Infinitely many distinct B_k contradict this finite dimension. The constants C'_k may vary; no bound on their supremum has been used. The exponent A , the weights, and the Sobolev orders were fixed throughout. This excludes the plane case and completes the proof. $\square$

7 Counting rational curves and obtaining the shell

We apply the preceding analysis to every datum of Lemma 3.1, using the decay rate fixed after Lemma 5.1. Proposition 4.11 gives a divisor D_i with ordinary class $j(e_i)$. By Theorem 6.6, every irreducible component of D_i is compact and has rational normalization. The divisor as a whole may still have infinitely many components; it is locally finite and bounded above in height. This distinction will be retained in all intersection pairings below.

Lemma 7.1 (Lifts of rational curves). *Every compact rational curve in Y projects to a rational curve in X . For each rational curve $B \subset X$, the compact irreducible curves in Y projecting onto B form one orbit under T . In particular, a collection of r rational curves in X gives at most nr orbits of compact lifts under $S = T^n$.*

Proof. The image under π of a compact curve is a compact analytic curve, because the map restricted to that curve is proper and locally finite. It is nonconstant since π is a local biholomorphism. If the upstairs normalization is $\mathbb{P}^1$, the induced nonconstant map to the normalization of the image is a finite map from $\mathbb{P}^1$ to a compact Riemann surface. Riemann–Hurwitz forces that surface to have genus zero. Thus the image is rational.

Conversely, let $\nu : \mathbb{P}^1 \rightarrow B \subset X$ be the normalization of a rational curve. Since $\mathbb{P}^1$ is simply connected, the map ν lifts to a holomorphic map $\tilde{\nu} : \mathbb{P}^1 \rightarrow Y$ after a choice of the image of one

point. Its image $\tilde{B}$ is compact and irreducible. Over the smooth part of B , its projection has degree one, because ν does. The lifts of this fixed normalization map form one orbit under the deck group, so their images do also.

To see that these are all the compact irreducible curves over B , take such a curve C and a point of it over the smooth part of B . Choose the lift of ν through this point. The local biholomorphism property of π makes its image and C agree in a nonempty open curve germ. Irreducibility and analytic continuation then make their images equal. This proves the orbit assertion. Splitting the integer deck powers into their n residue classes proves the bound for S .

This argument lifts the normalization map, not the singular topological space B . In particular it also applies to a nodal rational curve; the connected components of its full inverse image need not be compact. $\square$

One curve downstairs accounts for infinitely many translates upstairs. We therefore count independent compact classes over a ring in which multiplication by t records one deck translation. A single orbit of compact curves will then impose one linear equation over the fraction field of that ring. The intersection pairing will show that fewer curve orbits than independent classes are impossible.

Fix n , put $S = T^n$, and set

$$\Lambda = \mathbb{Q}[t, t^{-1}], \quad \mathbb{F} = \mathbb{Q}(t), \quad \overline{f(t)} = f(t^{-1}).$$

On compactly supported cohomology, let

$$ta = S_! a := (S^{-1})^* a.$$

This is the convention that translates the support in the positive S -direction. It agrees with the pushforward action $C \mapsto S_* C$ on homology under Poincaré duality. Define

$$P(a, b) = \sum_{k \in \mathbb{Z}} \langle a, S_!^k b \rangle t^k \quad (a, b \in H_c^2(Y; \mathbb{Q})). \quad (7.1)$$

This is a Laurent polynomial: compactly supported representatives intersect only finitely many translates of one another. Invariance of intersection under S and symmetry in degree two give

$$P(ta, b) = tP(a, b), \quad P(a, tb) = t^{-1}P(a, b), \quad P(b, a) = \overline{P(a, b)}. \quad (7.2)$$

Thus P is linear in its first variable and conjugate-linear in its second, where conjugation means only the involution $t \mapsto t^{-1}$. For a compact oriented curve C , define similarly

$$Q(a, C) = \sum_{k \in \mathbb{Z}} \left(\int_{S_*^k C} a \right) t^k. \quad (7.3)$$

This is finite and Λ -linear in a with the same convention. Its vanishing is equivalent to the vanishing of the ordinary pairing of a with every S -translate of C .

Lemma 7.2 (Nonsingularity of the compact classes). *For the classes $e_1, \dots, e_d$ of Lemma 3.1, the matrix $P(t) = (P(e_i, e_j))_{i,j=1}^d$ has nonzero determinant in Λ . Their Λ -span is free of rank d , and P is nonsingular on its extension of scalars to $\mathbb{F}$.*

Proof. Let w_i be the transferred classes in X_n . Periodizing compactly supported closed representatives of e_i and e_j , then integrating over one fundamental period, gives

$$P(e_i, e_j)(1) = \sum_{k \in \mathbb{Z}} \langle e_i, S_!^k e_j \rangle = w_i \cdot w_j. \quad (7.4)$$

For completeness, the periodization is the locally finite sum of all S_1 -translates of a representative; it descends to a closed form on X_n . In the integral of the wedge product of two such sums, change variables by the translate in the first factor. The translated fundamental periods tile Y , leaving the displayed sum indexed by the relative translate of the second factor. Compact supports justify every interchange. In the construction of Lemma 3.1 the forms are supported inside the same open period, so terms with $k \neq 0$ actually vanish; the periodization identity also records the intrinsic pairing convention.

The classes w_i are independent over $\mathbb{Q}$ and the form on X_n is negative definite. Their Gram matrix is therefore negative definite, in particular invertible. Equation (7.4) implies $\det P(1) \neq 0$, so $\det P(t)$ is not the zero Laurent polynomial. The matrix is invertible over $\mathbb{F}$. If $\sum_i f_i(t)e_i = 0$ for $f_i \in \Lambda$, pairing with each e_j and using (7.2) gives $(f_1, \dots, f_d)P(t) = 0$. Invertibility over $\mathbb{F}$ forces every $f_i = 0$. This proves freeness and the asserted nonsingularity. $\square$

Proposition 7.3 (The required number of rational curves). *The surface X contains at least $b_2(X)$ rational curves.*

Proof. If there are infinitely many, the conclusion is immediate. Otherwise suppose for a contradiction that their number is $r < b_2(X)$, and list all of them. Choose n so large that

$$n(b_2(X) - r) > C_0.$$

Lemma 3.1 then gives $d > nr$ compact classes. By Lemma 7.1, choose representatives $C_1, \dots, C_q$ for all S -orbits of compact irreducible lifts of the listed rational curves, where $q \leq nr < d$.

Consider over $\mathbb{F}$ the q homogeneous linear equations

$$\sum_{i=1}^d f_i(t) Q(e_i, C_\ell) = 0, \quad 1 \leq \ell \leq q. \quad (7.5)$$

There are more unknowns than equations, so a nonzero solution exists. Multiplying all its entries by a common nonzero denominator gives $f_i \in \Lambda$, not all zero. The compactly supported class

$$x = \sum_{i=1}^d f_i(t)e_i$$

is nonzero by Lemma 7.2; its support is compact because this is a finite Laurent combination. Equations (7.5) and (7.3) imply that x pairs to zero with every compact rational lift and every one of its S -translates.

For each j , the ordinary class $j(e_j)$ is represented by the divisor D_j . Every component of D_j is compact rational by Theorem 6.6, and its image downstairs is rational by Lemma 7.1. Hence each component occurs among the curves just annihilated by x . If α is a compactly supported closed real two-form representing x , the ordinary divisor identity (4.19) gives, for every $k \in \mathbb{Z}$,

$$\langle x, S_1^k e_j \rangle = \int_{S_*^k D_j} \alpha = 0.$$

Indeed local finiteness makes the last integral a finite sum of integrals over compact components meeting $\text{supp } \alpha$, and each summand is zero by the curve orthogonality. This argument pairs a compact test class with a locally finite divisor; it never pairs two locally finite cycles or sums an infinite Laurent series.

We have proved $P(x, e_j) = 0$ for all j . By (7.2) this says $(f_1, \dots, f_d)P(t) = 0$, contradicting the nonsingularity of $P(t)$ over $\mathbb{F}$. Therefore $r \geq b_2(X)$. $\square$

Lemma 7.4 (Independence of curve classes). *Under (2.2), the real cohomology classes of any finite collection of distinct curves on X are linearly independent. Consequently X has at most $b_2(X)$ curves in total.*

Proof. Suppose a finite collection admits a nonzero real relation. Split its positive and negative coefficients to obtain effective real divisors A and B with disjoint lists of components and $[A] = [B] = v$; one of them may be empty. If $v \neq 0$, both are nonempty and negative definiteness gives $v^2 < 0$. But positivity of local intersections of distinct complex curves gives

$$v^2 = A \cdot B \geq 0,$$

a contradiction. Thus $v = 0$. At least one nonempty effective real divisor, say $A = \sum_{i \in I} a_i B_i$ with all $a_i > 0$, has zero real class.

To pass from this real relation to an integral one, express the classes $[B_i]$ in an integral basis of $H^2(X; \mathbb{Z})/\text{Tors}$. The relation space is the kernel of an integer matrix, hence has a basis over $\mathbb{Q}$. Its rational points are dense in its real points. Approximate $(a_i)_{i \in I}$ within this kernel by a rational vector with all coordinates still positive, and clear denominators. The result is a nonzero effective integral divisor with zero real class and hence square zero. This contradicts (2.2). There is no relation. Since $\dim H^2(X; \mathbb{R}) = b_2(X)$, any collection of more than $b_2(X)$ curves would contain such a finite dependent collection. $\square$

Proof of Theorem 1.1. If X has a nonzero effective square-zero divisor, Proposition 2.1(i) already gives the desired shell. In the remaining case, Proposition 7.3 and Lemma 7.4 show that X has exactly $b_2(X)$ rational curves. It is compact and minimal, has $b_1(X) = 1$, $\kappa(X) = -\infty$, and $b_2(X) > 0$ by the original hypotheses. Every hypothesis of the Main Theorem of Dloussky–Oeljeklaus–Toma is therefore satisfied, so Proposition 2.1(ii) applies.

The conclusion in [11, pp. 283–284] is an open neighborhood U of the unit sphere in $\mathbb{C}^2 \setminus \{0\}$ and a biholomorphism $\phi : U \rightarrow \Sigma \subset X$ such that $X \setminus \Sigma$ is connected. This is precisely the conclusion required in Theorem 1.1. Equivalently, if a shell is described by its nonseparating core sphere, choose a sufficiently small round annular collar in the given holomorphic neighborhood. On either side of the core, the collar coordinate pushes the complement of the core onto the complement of the smaller open collar. The latter is therefore connected as well. $\square$

7.1 Consequences of the shell

The shell now gives the deformation and topological description stated in the introduction, and removes the remaining class VII exception in the aspherical-surface inequality.

Proof of Corollary 1.2. Theorem 1.1 supplies a global spherical shell. Kato’s deformation theorem [20, Theorem 1] then gives a one-parameter deformation with each nonzero fiber a modification of a primary Hopf surface; his Corollary 1 gives $\pi_1(X) \cong \mathbb{Z}$. In complex dimension two, a modification between smooth surfaces factors as a finite succession of point blowups, with possibly infinitely near centers [3].

After shrinking Δ , the family is a proper smooth submersion. Ehresmann’s theorem identifies its fibers by diffeomorphisms preserving their continuously varying complex orientations. Every primary Hopf surface is diffeomorphic to $S^1 \times S^3$ [16, §1, p. 1109], and hence has second Betti number zero. Each complex point blowup increases the second Betti number by one and replaces the oriented smooth manifold by its connected sum with $\overline{\mathbb{CP}}^2$ [3]. Since the fibers have second Betti number b , there are exactly b blowups, and the asserted oriented smooth type follows. The product orientation can be chosen compatibly because $S^1 \times S^3$ admits an orientation-reversing diffeomorphism. $\square$

Proof of Corollary 1.3. Albanese–Di Cerbo–Lombardi [1, Theorem 1.4] prove these conclusions except for the possible case of a class VII_0^+ surface violating the Global Spherical Shell conjecture. Theorem 1.1 excludes that case. Corollary 1.2 also explains the non-asphericity: such a surface has $\pi_1 \cong \mathbb{Z}$ and $b_2 > 0$, but an aspherical manifold with fundamental group $\mathbb{Z}$ is homotopy equivalent to S^1 . $\square$

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